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Transformer-based Expansion of Dutch Dictionary Definitions

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Preface

I would like to thank Professor Vincent Vandeghinste and Professor Kris Heylen for their guidance and for the confidence they showed in allowing me to pursue this master's thesis alongside my traineeship at the European Commission's Joint Research Centre in agentic AI. Balancing both commitments has broadened my academic perspective and meaningfully contributed to my professional growth.

Robbe Meersman

Contents

Abstract

The manual expansion of concise, core-meaning definitions into comprehensive, formally structured dictionary entries represents a significant bottleneck in contemporary lexicography, demanding substantial time and expertise. This research investigates the potential of transformer-based models to automate this sense-preserving definitional expansion for Dutch, a task that requires not only semantic accuracy but also strict adherence to lexicographical style and structure.

The study empirically compares two primary methodologies for leveraging large language models: in-context learning via few-shot prompting and adaptation via parameter-efficient fine-tuning with quantized low-rank adaptation. A range of powerful multilingual and Dutch-specific models, including mT5-xl, GEITje Ultra, Aya-101 and Aya-23 are evaluated. Model performance is assessed using a hybrid framework combining automated metrics for semantic accuracy with structured qualitative analysis to judge lexicographical appropriateness.

The results demonstrate that fine-tuning is a transformative and superior approach for this specialized domain. While few-shot prompting yields high quantitative scores with one model, qualitative analysis reveals this performance to be fundamentally unreliable, characterized by an inconsistent performance profile with outputs ranging from perfect replications to complete failures. This discrepancy arises because fine-tuning enables models to internalize and robustly generalize lexicographical patterns, whereas few-shot learning operates closer to brittle pattern mimicry, its success entirely contingent on the structural quality of the provided examples. The fine-tuned Aya-23 model emerges as the most effective overall, distinguished not only by high semantic accuracy scores but, more importantly, by its consistent reliability and strong adherence to lexicographical conventions.

The findings establish that for high-precision domains, task-specific fine-tuning is essential to overcome the stylistic priors of general-purpose models and instill the discipline required for generating reliable, domain-appropriate output. This work provides a viable methodology for developing a practical "lexicographer's assistant," contributing to a vision of a human-in-the-loop system where the model generates high-quality first drafts and the human expert's role evolves to that of a critical validator and refiner.

List of figures and tables

List of figures

List of tables

List of abbreviations

Abbreviations

ANW	Algemeen Nederlands Woordenboek
BERT	Bidirectional Encoder Representations from Transformers
DPO	Direct Preference Optimization
ELMo	Embeddings from Language Models
GloVe	Global Vectors for Word Representation
LLM	Large Language Model
LSTM	Long Short-Term Memory
LoRA	Low-Rank Adaptation
NLP	Natural Language Processing
PEFT	Parameter-Efficient Fine-Tuning
POS	Part-Of-Speech
QLoRA	Quantized Low-Rank Adaptation
RNN	Recurrent Neural Network
Seq2Seq	Sequence-to-Sequence
SFT	Supervised Fine-Tuning
WMT	Workshop on Machine Translation

1 Introduction

The practice of lexicography, the craft of compiling, writing and editing dictionaries, has historically been a meticulous and labour-intensive discipline. The creation of high-quality dictionary entries is a specialized skill that demands not only deep linguistic knowledge but also a significant investment of time and human effort. For comprehensive, large-scale dictionary projects, this process can span many years, representing a substantial undertaking in preserving and describing a language. It is within this context of immense resource requirements that the field of computational lexicography has emerged, seeking to apply techniques from natural language processing (NLP) to assist in and, where possible, automate aspects of the lexicographical workflow.

1.1 Background

A central task within the domain of NLP is that of definition modelling, a term formally introduced by Noraset et al. (2017) to describe the automatic generation of a natural language definition for a given word [1]. The original motivation for this task was not merely text generation as an end in itself. Rather, it was proposed as a more direct and transparent method for evaluating the semantic richness encoded within word embeddings. The core premise was that a high-quality, semantically potent vector representation of a word should contain sufficient information to reconstruct its corresponding dictionary definition. Early approaches to this task employed Recurrent Neural Network (RNN) architectures to generate definitions from static word vectors. These initial studies revealed a persistent challenge in the field: how to effectively condition the generation process on the word being defined. A simple 'seed' model, where the word vector was provided only as the initial input to the RNN, was found to be significantly outperformed by a 'gated' model, which used a gating mechanism to control the flow of information from the target word's embedding to the generation process at each time step. This finding underscored the difficulty of maintaining focus on the target concept while generating a fluent and coherent definition.

A more fundamental limitation of these early models was their inability to handle polysemy, the common linguistic phenomenon where a single word form possesses multiple distinct meanings. A model conditioned on a single, static vector for a word like 'bank' would be incapable of distinguishing between its financial and geographical senses, likely defaulting to the meaning most frequent in its training data. This critical issue marked a turning point in the research trajectory and was explicitly addressed by Gadetsky et al. (2018) [2]. They proposed a conditional generation framework where the model was provided not only with the word to be defined but also with an example sentence that supplied disambiguating context. This innovation effectively shifted the task from word-level definition modelling to the more nuanced and practical task of sense-level definition generation.

This evolution from word-level to sense-level generation provides the direct scholarly context for the present research. More recently, the field of NLP has been reshaped by the introduction of the transformer architecture and the subsequent development of Large Language Models (LLMs) [3]. These models have demonstrated powerful text generation capabilities and recent studies have begun to explore their potential for lexicographical tasks [4]. This thesis builds on these advancements. The task investigated represents a logical progression in the line of inquiry established by earlier definition modelling research, but it leverages these modern neural architectures.

Specifically, a transformer-based model is provided with a lemma and a concise, core-meaning definition (a short definition) and is tasked with generating a more comprehensive, formal definition (a full definition). The input is therefore not a sentential context but a definitional one. Consequently, the primary challenge is not word sense disambiguation, as the intended sense is already specified by the short definition. Instead, the task is more accurately described as sense-preserving definitional expansion. The model must learn to capture the core semantics of the concise input definition and elaborate upon it to produce a full, lexicographically appropriate definition, without drifting to another sense of the same lemma. This framing positions the project to build upon the foundations of context-aware definition modelling, applying contemporary generative techniques to a novel form of definitional input.

1.2 Problem statement

The motivation for this research is grounded in the workflow of contemporary lexicographical projects such as the Algemeen Nederlands Woordenboek (ANW). This is a dictionary of modern Dutch provided by the Dutch Language Institute, from which this study's dataset is derived. A significant portion of the manual effort in such projects is dedicated to the expansion of core definitions into the detailed, formally structured and stylistically consistent entries required for a dictionary. This process of expansion is a well-recognised bottleneck, demanding substantial time and expertise from trained lexicographers. This thesis investigates the potential of modern transformer technology to alleviate this bottleneck by automating the generation of these expanded definitions for Dutch.

The technological foundation for this investigation is the paradigm shift in NLP brought about by the advent of large-scale transformer models and the unified text-to-text framework. As comprehensively explored by Raffel et al. (2020) in their work on the text-to-text transfer transformer (T5), this framework posits that nearly any NLP task can be cast as a problem of transforming an input text sequence into an output text sequence [5]. This approach offers considerable advantages by allowing the same model architecture, objective function and decoding process to be applied across a diverse range of tasks, thereby simplifying the experimental setup and effectively leveraging the power of transfer learning from models pre-trained on vast corpora.

The timeliness of this research is underscored by recent studies exploring the capabilities and limitations of LLMs in the domain of lexicography. A key motivating study by Lew (2023) investigated the use of ChatGPT for generating COBUILD-style dictionary entries and provided a nuanced assessment of its performance [4]. The findings revealed a crucial duality in the model's capabilities: while the generated definitions were judged by human experts to be of a quality comparable to those produced by professional lexicographers, the LLM generated example sentences and the overall entries received significantly lower ratings. This suggests that while modern LLMs possess strong semantic competence, they exhibit weaknesses in adhering to the specific stylistic conventions, structural requirements and authentic phrasing characteristic of high-quality lexicographical work.

This duality defines the central problem addressed in this thesis. The challenge is not merely to achieve semantic expansion of a concise definition but to do so in a manner that is stylistically and structurally appropriate for a dictionary. This problem is particularly pronounced when considering the use of models that have been primarily tuned for conversational or general-purpose instruction following, which may create a domain and style mismatch with the formal, structured nature of lexicographical text. The research will be conducted in the context of Dutch, a language for which dedicated large-scale generative models are less common than for English. This necessitates the use of large multilingual models, which in turn makes full fine-tuning computationally prohibitive on typical academic hardware. To address this, the research will employ Parameter-Efficient Fine-Tuning (PEFT).

1.3 Research questions

Based on the background and problem statement, this thesis is guided by three central research questions designed to systematically investigate the potential of transformer-based models for Dutch definition expansion.

1. *How effectively can modern LLMs generate lexicographically sound expansions of Dutch definitions using only few-shot prompting?*

This first question aims to establish a robust baseline by evaluating the in-context learning capabilities of a powerful, general-purpose Dutch language model. Inspired by the methodology of Lew (2023), this approach simulates a scenario where limited or no parallel training data is available, a common situation in practical lexicography [4]. It seeks to determine the out-of-the-box performance of LLMs on this specialized task.

2. *How does the performance of parameter-efficient fine-tuned LLMs compare to the few-shot prompting baseline in terms of semantic accuracy and stylistic appropriateness?*

This is the core empirical question of the thesis. It directly compares the baseline performance against that of models specifically adapted to the task. The goal is to determine whether specialized training can overcome the stylistic and structural weaknesses observed in general-purpose LLMs and provide a superior solution for generating high-quality lexicographical text. This question explicitly tests the hypothesis that fine-tuning can bridge the domain mismatch between general-purpose models and the target lexicographical style.

3. *What constitutes an effective hybrid framework for evaluating both the semantic accuracy and lexicographical quality of generated definitions?*

This question addresses the inherent complexity of evaluating the generated output. Recognising that the quality of a dictionary definition is multi-faceted, this question motivates the development of a hybrid evaluation protocol. Given the known limitations of traditional lexical overlap metrics like BLEU for assessing the nuanced output of LLMs [6], this research will explore a combination of modern, embedding-based neural metrics such as BERTScore and COMET to assess semantic accuracy, alongside a structured qualitative error analysis to evaluate lexicographical appropriateness, including stylistic and formal correctness.

1.4 Contributions and significance

This research is expected to make several contributions to the field of computational lexicography and natural language processing. The primary contributions are:

1. The development of several transformer models adapted using the QLoRA technique, which are specifically fine-tuned for the task of expanding Dutch dictionary definitions. As a direct response to the limited availability of specialized resources for Dutch computational lexicography, these models will represent a new, specialized resource for the community.
2. A systematic empirical comparison between a fine-tuning-free, few-shot prompting approach and a specialized, parameter-efficient fine-tuning approach. This comparison will provide guidance on the most effective strategy for applying LLMs to this specific lexicographical task.
3. The design and application of a multi-faceted evaluation protocol that combines state-of-the-art automated metrics with a structured, linguistically informed qualitative error analysis.

The significance of this work extends beyond its specific contributions. The research is not aimed at demonstrating that LLMs can fully replace the human lexicographer. Rather, its true significance lies in its potential to create a practical and effective lexicographer's assistant. By investigating how modern NLP models can best be leveraged for definition expansion, this thesis aims to pave the way for tools that can generate high-quality first drafts of dictionary entries. Such a system has the potential to significantly reduce the manual effort, time and cost associated with dictionary production, improve the consistency of entries across a large resource and ultimately accelerate the creation of valuable linguistic resources like the ANW. This positions the research as a direct and practical contribution to the evolving practice of modern lexicography.

1.5 Thesis outline

The remainder of this thesis is structured into five more chapters to systematically address the research questions and present the findings in a clear and logical manner.

Chapter 2: Literature Review
The second chapter provides a comprehensive review of the relevant literature. It begins by tracing the history of definition modeling, then delves into the transformer-based architectures and fine-tuning techniques that form the technical foundation of

this work. Finally, it details the hybrid framework developed for evaluating the generated results.

Chapter 3: Methodology
The third chapter outlines the research methodology. This includes a thorough description of the dataset used, a step-by-step account of the experimental setup for both the few-shot baseline and the fine-tuned models and a detailed explanation of the evaluation process.

Chapter 4: Results
The fourth chapter presents the empirical findings of the experiments. It offers a quantitative analysis through detailed metrics and complements this with a qualitative analysis of the expanded definitions, highlighting key patterns and outcomes.

Chapter 5: Discussion
The fifth chapter is dedicated to a discussion of the results. Here, the findings are interpreted to directly answer the research questions posed in the introduction. The chapter focuses on the effectiveness of the different techniques employed, drawing conclusions from the experimental outcomes.

Chapter 6: Conclusion
Finally, the sixth chapter concludes the thesis. It provides a summary of the main contributions and findings, offers a critical reflection on the study's limitations and proposes promising directions for future research.

2 Literature review

This chapter establishes the scholarly foundation for the thesis by reviewing the key technological and conceptual developments that inform the task of definition expansion. The review begins with a historical analysis of automatic definition generation. This historical context is crucial because it reveals that the central challenge of the field has long been the problem of polysemy. The inability of early models to handle a word's multiple meanings rendered reliable automation unfeasible, making the recent advent of context-aware transformers a critical turning point that makes the task newly viable and worthy of investigation. Understanding this long-standing limitation is what frames the current research not as simple text generation, but as a targeted investigation into sense-preserving expansion. This historical context is therefore essential, as it clarifies the technical hurdles that have long defined the field and provides the necessary backdrop for discussing the modern solution this thesis investigates. The review then proceeds to survey the landscape of contemporary foundation models suitable for Dutch, justifying the selection of specific candidates for this study. Finally, the review covers the essential practical considerations for the research, outlining the parameter-efficient fine-tuning methods required to adapt these large models and the hybrid evaluation framework needed to assess both the semantic accuracy and the lexicographical quality of the generated output.

2.1 Early approaches to definition modelling

2.1.1 The foundational symbolic paradigm: WordNet

The cornerstone of modern computational lexicography is arguably WordNet, a project initiated in 1985 at Princeton University. Unlike traditional dictionaries organised by alphabetical word forms, WordNet was designed to be a lexical reference system whose structure is "inspired by, what was then, the current psycholinguistic theories of human lexical memory" [7]. Its fundamental organising principle is not the word but the concept, represented by a set of cognitive synonyms, or a "synset". Each synset contains a brief definition, or "gloss," and is linked to other synsets through a network of explicit semantic relations, such as hyponymy (is-a), hypernymy (is-a-kind-of) and meronymy (is-a-part-of) [7, 8].

From its earliest project proposals such as Miller (1995) [9] and the more detailed introductory report by Miller et al. (1993) [7], WordNet was conceived both as a meaning-centred alternative to alphabetical dictionaries and the beginnings of machine-readable resource [8]. It established what a structured lexical knowledge base should contain: distinct word senses, human-readable definitions and a formal network of semantic relationships. However, its construction was a labour-intensive process requiring extensive human effort from linguists and lexicographers. This inherent limitation in scalability and coverage became the primary motivation for the next wave of research, which sought to automate the creation and expansion of such resources.

2.1.2 Early automation efforts

Recognizing the immense cost and effort involved in manually building resources like WordNet, researchers turned their attention to methods for automatic expansion. A representative example of this line of inquiry is the work of H. G. Oliveira and P. Gomes (2010), which explicitly aimed "towards the automatic creation of a wordnet from a term-based lexical network" [10]. Their work exemplifies the transition from purely handcrafted resources to hybrid and data-driven approaches.

The methodology described is characteristic of the symbolic and early statistical era. It involved a multi-step process: first, a clustering procedure was applied to a network of known synonyms to automatically derive candidate synsets. Second, relational triples (e.g., *term A - relation - term B*) extracted from other sources were mapped onto these newly formed synsets to build out the relational network. This approach frames the problem as a word sense disambiguation task where the only available information is the graph structure of the lexical network itself, rather than rich, sentential context.

This research effort highlights the field's awareness of the critical need for scalability and broader coverage, which were the primary bottlenecks of the manual lexicography paradigm. While these rule-based and graph-based methods were innovative, they were often brittle, struggled with the noise and ambiguity inherent in large-scale text and achieved lower precision than their handcrafted counterparts [10]. This struggle for a robust, scalable method for capturing meaning from data set the stage perfectly for the distributional semantics revolution.

2.2 Distributional semantics and word-level definition modelling

The limitations of symbolic and rule-based systems created an opening for a fundamentally different approach to meaning. The second major era in the journey towards automatic definition generation was driven by the distributional hypothesis: the idea that words occurring in similar contexts tend to have similar meanings [11]. This section traces the operationalisation of this hypothesis through the development of dense word embeddings and marks the first successful attempt to bridge the new neural paradigm with the traditional task of definition modelling.

2.2.1 The rise of dense word embeddings

The theoretical foundation of distributional semantics was translated into a powerful computational reality with the introduction of word embeddings. These are dense, low-dimensional vector representations of words learned from large text corpora. The most influential work in this area was introduced by Mikolov et al. in two seminal papers in 2013. The first, "Efficient Estimation of Word Representations in Vector Space," introduced the word2vec models: Continuous Bag-of-Words (CBOW) and Skip-gram [12]. These models offered a radical simplification of previous neural language models. Instead of complex architectures with non-linear hidden layers, word2vec employed simple, computationally efficient log-linear models that either predicted a target word from its surrounding context words (CBOW) or, conversely, predicted the context words from a target word (Skip-gram). This efficiency allowed them to be trained on billions of words of text, an unprecedented scale at the time. The follow-up paper, "Distributed Representations of Words and Phrases and their Compositionality," introduced further crucial optimizations like negative sampling and subsampling of frequent words, which accelerated training and improved vector quality [13]. Crucially, this work demonstrated that the resulting vector spaces captured not just broad similarity but also "a large number of precise syntactic and semantic word relationships" [13] that could be expressed as simple vector arithmetic. (e.g. $vector(king) - vector(man) + vector(woman) \approx vector(queen)$)

Appearing shortly after, GloVe (Global Vectors for Word Representation) by Pennington, Socher and Manning (2014) offered an alternative yet complementary approach [14]. GloVe was explicitly designed to unify the two dominant modeling families: local context window methods like word2vec and global matrix factorization methods like Latent Semantic Analysis (LSA). The core intuition of GloVe is that the ratios of word-word co-occurrence probabilities hold the key to meaning. For instance, the ratio of probabilities $P(solid|ice)/P(gas|ice)$ will be high, while

$P(solid|steam)/P(gas|steam)$ will be low. The model is trained via a weighted least-squares objective to produce vectors whose dot product equals the logarithm of their co-occurrence probability, effectively encoding these meaningful ratios as vector differences.

2.2.2 From word representation to definition generation

These new word representations provided the necessary toolkit for the next conceptual leap: using neural networks not just to represent words, but to model the relationship between a word and its definition. The pivotal work here is "Learning to Understand Phrases by Embedding the Dictionary" by Hill et al. (2016) [15]. This paper bridges the symbolic world of traditional lexicography with the emerging neural paradigm.

The authors proposed a novel training task that directly operationalized the concept of a definition. They used a Recurrent Neural Network (RNN), specifically a Long Short-Term Memory (LSTM) network, to process a definition (e.g., "a tall, long-necked, spotted ruminant of Africa") as a sequence of word embeddings. The model's objective was to learn a mapping from this input sequence to a single output vector: the pre-trained word2vec embedding of the word being defined, in this case, "giraffe". The model's success on practical tasks like building a "reverse dictionary" (finding a word for a given description) and solving crossword clues demonstrated that the compositional meaning of a phrasal definition could be effectively encoded into a vector that aligns with the distributional semantics of its target word. This work represents a first clear instance of generative "definition modelling" in the neural era and provided a path from representing single words to understanding phrases.

2.2.3 The inherent limitation of static representations

The very success of static word embeddings like word2vec and GloVe illuminated their biggest weakness: an inability to model polysemy, the phenomenon where a single word form carries multiple distinct meanings. By design, these models assign one unique, context-independent vector to each word type in the vocabulary. The vector for "bank," for example, is necessarily a conflation of all its senses (financial institution, river's edge, a turn in aviation, etc.) derived from an amalgamation of all the different contexts in which "bank" appeared in the training corpus. This is not a flaw that can be fixed with more data or a larger model; it is a fundamental architectural limitation [16].

This shortcoming of static embeddings is inherited directly by the definition modeling approach of Hill et al. (2016) [15]. Their model learns to map the definition "a financial institution" to the single, averaged vector for the word "bank." While this works as a retrieval task due to the strong signal in the training data, the model is incapable of representing the fact that this definition corresponds to a different sense of "bank" than a definition like "the sloping side of a river." It cannot disentangle the multiple meanings packed into that single vector.

Thus, the adoption of static embeddings established the central research challenge for the next generation of models. It became clear that the field required dynamic and context-sensitive representations, rather than static embeddings. Specifically, models needed to generate distinct embeddings for words such as "bank" depending on whether they occurred in financial or geographical contexts. This necessity for genuine contextualization directly motivated the development of models such as ELMo and BERT [17, 18], which subsequently defined the next research paradigm.

2.3 Self-attention and the transformer revolution

The evolution of sequence modelling reached its most decisive inflection point with the development of the transformer architecture and the paradigm of large-scale contextual pre-training. This shift was built upon a critical preceding innovation. In machine translation, standard RNNs faced a "bottleneck" where they struggled to compress long sentences into a single fixed-length vector. Bahdanau et al. (2015) introduced the attention mechanism to solve this, allowing a model to selectively focus on relevant parts of the input when generating output [19]. This powerful idea led to the next great leap: an architecture that dispensed with recurrence entirely.

2.3.1 The architectural breakthrough: the transformer

The paper that catalysed this shift was "Attention Is All You Need" by Vaswani et al. (2017) [3]. The authors proposed the transformer, a novel architecture that forwent recurrence and convolutions entirely, relying solely on attention mechanisms to model dependencies between input and output. The central innovation was self-attention, also known as intra-attention. Unlike the attention mechanism of Bahdanau et al. (2015), which related positions in a target sequence to positions in a source sequence [19], self-attention relates different positions within a single sequence to compute a new representation of that same sequence. In essence, for each word, self-attention calculates a weighted average of all other words in the same sequence, where the weights determine how relevant each word is to the current one. This architectural choice had profound consequences. By eliminating the sequential computations inherent in RNNs, the transformer could process all tokens in an input sequence simultaneously. This massive parallelisation dramatically reduced training times and, critically, made it feasible to train models on a scale previously unimaginable.

2.3.2 The semantic breakthrough: contextual embeddings

While the transformer provided a more powerful and scalable architecture, the solution to the polysemy problem came from a new approach to pre-training. Peters et al. (2018) introduced Embeddings from Language Models (ELMo), a model that generated word representations as a "function of the entire input sentence" [17]. ELMo used a deep bidirectional Long Short-Term Memory (biLSTM) network pre-trained on a large text corpus with a language modelling objective. For any given downstream task, the representation for a token was not a fixed vector looked up in a table, but a dynamically computed, task-specific linear combination of all the internal layer representations of the pre-trained bidirectional language model (biLM) architecture. This meant the vector for "play" in "a broadway play" would be different from the vector for "play" in "the children play," effectively solving the static embedding problem that had plagued word2vec and GloVe [12, 13, 14]. ELMo's success in significantly improving the state of the art across several challenging NLP tasks (e.g. including textual entailment, question answering and sentiment analysis) proved the immense value of transferring knowledge from deep, contextual models pre-trained on general-purpose language modelling.

2.3.3 The synthesis: BERT

The synthesis of the transformer architecture and contextual pre-training culminated in the introduction of Bidirectional Encoder Representations from transformers (BERT) by Devlin et al. (2019) [18]. BERT replaced the LSTM-based approach of ELMo with the transformer, enhancing both representational power and computational efficiency. Crucially, BERT introduced a novel pre-training strategy capable of achieving deep bidirectionality in language representations, overcoming limitations inherent in earlier models.

ELMo achieved bidirectionality by separately training forward and backward LSTMs, subsequently concatenating their outputs. In contrast, BERT implemented the masked language model objective, randomly masking tokens in the input sequence. During training, the model predicted these masked tokens based on the complete left and right context simultaneously [18]. This masked prediction method enabled BERT to create deep, genuinely bidirectional representations conditioned on the entire sentence context, something unattainable by unidirectional approaches or shallow concatenation methods such as those employed by ELMo.

The introduction of BERT represented a breakthrough, achieving state-of-the-art results on a wide range of NLP benchmarks, including GLUE (General Language Understanding Evaluation) and SQuAD (Stanford Question Answering Dataset), often surpassing previous baselines by large margins [20]. Its success solidified the "pre-train and fine-tune" approach as the prevailing paradigm in NLP, firmly establishing the transformer encoder architecture as foundational for subsequent developments in language understanding tasks [21].

The evolution leading to BERT exemplified a mutually reinforcing dynamic between model architecture, training objectives and computational infrastructure. The parallelisable nature of the transformer architecture enabled training at scales previously unattainable with sequential models such as LSTMs. Concurrently, the impressive empirical outcomes of contextual pre-training first demonstrated by ELMo, and subsequently improved by BERT, justified the extensive investment in computational and engineering resources. This created a virtuous cycle wherein more scalable architectures allowed training on larger datasets, unveiling new emergent model capabilities, which subsequently motivated the development of even larger models and further hardware optimisation [21].

2.4 The text-to-text framework

Following the establishment of the pre-trained transformer as the dominant architecture, the next significant shift was not architectural but conceptual. Researchers sought to simplify and unify the way these powerful models were applied to the diverse landscape of NLP tasks. This led to the text-to-text framework, a paradigm that recasts every task as a text generation problem. This move effectively relocated the locus of research complexity from the design of model architectures to the design of model inputs, or "prompts," enabling a more flexible and unified approach to tasks like definition generation.

2.4.1 The unifying framework: T5

Raffel et al. (2020) canonised this approach with the T5 (text-to-text transfer transformer) model. The core philosophy of T5 is to treat every NLP problem as a "text-to-text" task, where the model takes a text string as input and is trained to produce a text string as output [5].

This framework provides a simple yet powerful unification. For example:

- **Machine Translation:** The input is prefixed with a task description, such as *translate English to German: That is good.*, and the model's target output is the string *Das ist gut*.
- **Text Classification:** A sentence is prefixed, e.g., *sst2 sentence: That is a good movie.*, and the model is trained to generate the literal string of the class label, *positive*.
- **Summarisation:** The input is prefixed with *summarise: [article text]*, and the target is the summary.

This approach allows the exact same encoder-decoder model, loss function and training procedure to be used across this diverse set of tasks, with the task itself being specified by the textual prefix in the input. Using a massive new dataset, the "Colossal Clean Crawled Corpus" (C4), the T5 model was pre-trained on a novel "span corruption" objective, which is a variation of BERT's masked language modelling task. The key difference is that while BERT's masked language modelling hides and predicts individual words, T5's span corruption objective replaces a random sequence of multiple words with a single special token and trains the model to regenerate the entire missing sequence.

2.4.2 Prompting and in-context learning

A groundbreaking shift occurred with the realisation that LLMs could be guided to perform tasks without any changes to their underlying parameters. The seminal paper by Brown et al. (2020), "Language Models are Few-Shot Learners," was instrumental in popularising this approach [22]. The authors demonstrated that their model, GPT-3, could perform a novel task based solely on a natural language description and a few examples provided within its input prompt. This ability, termed in-context learning, allows the model to "learn" from the provided demonstrations without any updates to the model's weights. Brown et al. systematically evaluated this capability under three distinct conditions:

- **Zero-shot learning:** The model is given only a natural language instruction describing the task (e.g., "Expand the following short definition into a full one:"). It must perform the task without seeing any examples, relying entirely on the vast knowledge acquired during its pre-training phase.
- **One-shot learning:** The model is provided with a single demonstration of the task (one input-output pair) in addition to the instruction.
- **Few-shot learning:** The model is provided with multiple demonstrations, typically as many as can fit into the model's context window.

The results showed that performance consistently improved with the number of demonstrations provided. Few-shot learning often achieved results competitive with, or even superior to, fine-tuned models across a variety of NLP tasks. This established in-context learning as a viable alternative to fine-tuning, providing the scholarly foundation for this thesis's first research question. This question seeks to determine the extent to which these few-shot prompting techniques can be applied to the specialized task of Dutch definition expansion without recourse to task-specific fine-tuning.

2.5 Foundation models for Dutch

This section goes over into the state-of-the-art in multilingual foundation models relevant to Dutch, providing an analysis of their architectures, training data and fine-tune methods. The goal is to establish a clear rationale for the selection of candidate models for the definition expansion task, grounding the choice in the specific strengths and weaknesses of each architecture.

2.5.1 Encoder-decoder: the case of mT5

The multilingual text-to-text transfer transformer (mT5) model represents a direct application of the text-to-text paradigm to a multilingual context [23]. As a multilingual variant of T5, its encoder-decoder architecture is inherently structured for conditional generation tasks. This makes it a natural fit for definition expansion, where the input is a concise definition and the desired output is a longer, more descriptive one.

A critical factor in its suitability is the composition of its pre-training data, the multilingual Colossal Clean Crawled Corpus (mC4), a vast dataset derived from Common Crawl that encompasses 101 languages [23]. The original paper by Xue et al. (2021) details the scale of the Dutch data used. The Dutch component of mC4 comprises 73 billion tokens (words or part of words). To put this number in context against the entire dataset, the mC4 corpus contains 6.3 trillion tokens in total. Therefore, while substantial, the Dutch data constitutes only about 1.16% of the raw corpus. Raw web-crawled data is naturally imbalanced, with languages like English appearing far more frequently. To mitigate this, the pre-training process employed a temperature-based sampling strategy. This technique deliberately up sampled the Dutch data, boosting its presence in the training mix to 1.98% of the total training examples, a proportion higher than its raw representation in the corpus would suggest [23]. Crucially, mT5 was pre-trained exclusively on the unsupervised, masked language modelling objective span corruption. This means the model learned to predict missing spans of text from raw linguistic data and was not subjected to any supervised instruction or chat-based fine-tuning. Consequently, it must be adapted to any specific downstream task through fine-tuning, making it not suitable for few-shot prompting.

The purely unsupervised pre-training of mT5 presents a compelling case for its inclusion as a baseline. Unlike models that have been heavily aligned for conversational interaction, mT5 offers a blank slate for domain adaptation. It has learned the deep statistical properties of the Dutch language from a massive dataset but has not been biased towards a particular interaction style. Lexicographical text is highly structured, formal and non-conversational, demanding precision and a specific encyclopedic tone.

2.5.2 Decoder-only: GEITje Ultra and the domain mismatch hypothesis

In contrast to the general-purpose pre-training of mT5, GEITje Ultra 7B is a state-of-the-art model specifically tailored for the Dutch language [24]. The original GEITje is built upon the Mistral 7B decoder-only transformer architecture and underwent a full-parameter finetune on 10 billion tokens of Dutch text sourced from the Dutch Gigacorpus and the MADLAD-400 web crawl [25]. This extensive, Dutch-centric training phase establishes it as a formidable candidate for any Dutch NLP task.

GEITje was further subjected to an alignment process resulting in GEITje Ultra 7B and this introduces a critical consideration interesting for comparing models. GEITje Ultra 7B was first subjected to Supervised Fine-Tuning (SFT) on a collection of synthetic, Dutch conversational datasets. This includes ultrachat_200k_dutch, alpaca-cleaned-dutch and dolly-15k-dutch, which were generated using gpt-3.5-turbo and gpt-4-turbo. This SFT stage, which resulted in the GEITje-7B-ultra-sft model, was designed to teach the model to follow instructions and engage in dialogue. Following SFT, the second stage involved further alignment using Direct Preference Optimization (DPO), a technique that refines a model based on preference data. This involved a synthetic feedback dataset where the model was trained to prefer outputs from gpt-4-turbo over those from an earlier version, GEITje Chat.

This extensive alignment towards a conversational, assistant-like persona gives rise to the domain mismatch hypothesis. The objective of DPO is to steer the model's outputs to more closely resemble the chosen responses in a preference dataset [26]. In GEITje's case, these preferred responses are helpful and conversational, not lexicographical. Consequently, the model's internal stylistic bias is heavily skewed towards this conversational style. When prompted to perform a formal task like definition expansion, it is plausible that the model will apply its learned conversational style, resulting in outputs that may be overly verbose, informal or structurally inconsistent with the concise and objective tone required for dictionary definitions. This potential mismatch presents a key challenge and a valuable point of comparison against the unaligned mT5.

2.5.3 Balancing breadth and depth: the Aya family

The Aya family of models offers an opportunity to explore the trade-offs between breadth and depth in multilingual pre-training. Aya-101 is a 13 billion parameter, instruction-tuned model that covers 101 languages [27]. It is architecturally based on mT5-xxl encoder-decoder architecture and was fine-tuned on a diverse collection of instruction datasets, including xP3x, the Aya Dataset, ShareGPT-Command and various translated datasets [28]. While its extensive language coverage is impressive, it risks suffering from the curse of multilingualism, where model capacity is distributed across so many languages that performance in any single language is diluted.

Aya-23 was developed as a direct response to this challenge. Available in 8 billion and 35 billion parameter versions, it focuses on a more concentrated set of 23 languages, including Dutch [29]. This model family is based on the high-performing Cohere Command series, which uses a decoder-only transformer architecture with optimizations such as parallel attention and feed-forward layers, SwiGLU activation, and Rotary Positional Embeddings (RoPE) to improve training efficiency and performance. The fine-tuning data for Aya-23 was meticulously curated, incorporating multilingual templates, human annotations from the Aya Collection, and, notably, synthetic data generated natively in all 23 target languages using Cohere's advanced Command R+ model. This ensures a high-quality, instruction-following dataset that is specifically tailored to the model's supported languages. This strategic decision to allocate more model capacity to fewer languages is intended to yield greater depth and superior performance for the covered languages.

2.5.4 The broader Dutch NLP landscape

To present a comprehensive view of the Dutch NLP landscape, it is important to acknowledge other significant models and initiatives. RobBERT, particularly its 2023 release, is a RoBERTa-based model that has been a state-of-the-art resource for Dutch for several years, trained on extensive Dutch web corpora like OSCAR [30]. As an encoder-only model, it is not suitable for generative tasks like definition expansion, but its existence underscores the availability of high-quality Dutch pre-trained models.

Looking forward, the GPT-NL project represents a major national initiative. Backed by the Dutch government and developed by organisations including TNO (innovation engine of the Netherlands) and the Netherlands Forensic Institute, GPT-NL aims to create an open and transparent foundation model that aligns with Dutch and European legislation and values [31]. While not yet available for use, its development signals a strong commitment to creating a robust and independent AI ecosystem for the Dutch language.

In addition to Dutch-specific initiatives, broader multilingual projects such as the ETH Zurich–EPFL collaboration also offer valuable resources for Dutch NLP. Scheduled for release in late summer 2025, this multilingual foundation model supports over 1,000 languages, though its training data primarily consists of English (approximately 60%) and non-English languages (over 1500), supplemented by code and mathematics data. It will be released under an open licence and made freely accessible, including complete transparency regarding the composition of its training data. The model is available in two sizes (8 billion and 70 billion parameters), allowing flexibility for different downstream tasks. Its transparent architecture (trained on the Swiss "Alps" supercomputer) and permissive licensing make it highly suitable for fine-tuning towards specific linguistic tasks, including Dutch lexicographical applications. Although not specifically trained on extensive Dutch corpora, the ETH–EPFL model provides a valuable baseline for exploring transfer learning and multilingual fine-tuning strategies. Its emphasis on openness, transparency and regulatory compliance aligns well with the goals of creating trustworthy, linguistically diverse AI models within a European context [32].

2.5.5 Model selection rationale

The selection of foundation models for this thesis is guided by a systematic evaluation of their architectural and training characteristics, with each model chosen to test a specific hypothesis relevant to the task of Dutch definition expansion. The chosen models (mT5, GEITje Ultra, Aya-101 and Aya-23) create a matrix of comparisons across several key dichotomies.

First, the selection allows for a direct comparison of encoder-decoder (mT5, Aya-101) versus decoder-only (GEITje, Aya-23) architectures. While encoder-decoder models are theoretically well-suited for seq2seq tasks, the impressive performance of modern decoder-only models warrants an empirical investigation into which paradigm is more effective for this specific conditional generation task of expanding dictionary definitions.

Second, the models represent a spectrum of training and alignment philosophies, enabling a comparison of the unsupervised pre-training (mT5), instruction alignment

(Aya-101 and Aya-23) and preference alignment (GEITje) approaches. This addresses the blank slate and domain mismatch hypotheses by asking whether a model with no stylistic priors adapts more readily to the formal lexicographical domain, a model pre-aligned for conversational use or instruction aligned models.

Third, the models embody different strategies for achieving proficiency in Dutch: massive multilingualism (Aya-101), focused multilingualism (Aya-23) and language-specific continued pre-training (GEITje). This comparison will shed light on the most effective path to developing high-quality expanded Dutch dictionary definitions.

2.6 Methodologies for fine-tuning LLMs

The sheer scale of modern foundation models presents a significant barrier to adaptation. A full fine-tuning approach, which involves updating all of the model's billions of parameters, is computationally prohibitive for most research settings. This computational and memory burden necessitates the use of Parameter-Efficient Fine-Tuning (PEFT) methods. These techniques adapt only a small fraction of the model's total parameters while keeping the majority of the pre-trained weights frozen, offering a practical path to model specialisation [33].

2.6.1 A reparameterization-based method

Among the various PEFT techniques, Low-Rank Adaptation (LoRA) has emerged as a particularly effective and widely adopted method. LoRA is based on the empirical observation that the weight changes a model needs during adaptation lie in a space of very low rank [34]. Instead of updating the entire pre-trained weight matrix W, LoRA freezes W and injects two smaller, trainable matrices, A and B. These matrices are designed such that their product has the same dimensions as the weight matrix, but the rank of the decomposition is much smaller than the original dimensions. The forward pass is then modified to compute $h=(W+BA)x$. The forward pass is the step in which the network turns an input x into an activation h. Because the weight matrix never changes, only the smaller matrices are updated during training. Freezing the weight matrix preserves the knowledge captured during pre-training while keeping the optimisation problem small [34].

This approach yields several significant benefits. First, it dramatically reduces the number of trainable parameters, often by over 99%, which in turn lowers the VRAM requirements for storing optimizer states and gradients. Second, the resulting adapter weights are small (often only a few megabytes), making it easy to store and manage many task-specific adapters for a single base model. Finally, because the update matrix BA can be merged with the original weight matrix W after training, LoRA introduces no additional inference latency, a key advantage for deployment [34].

2.6.2 Pushing the efficiency frontier: quantization and QLoRA

To further reduce the memory footprint of fine-tuning, LoRA can be combined with quantization. Quantization is a model compression technique that reduces the numerical precision of a model's weights, for example, from 16-bit floating-point numbers to 4-bit integers [35]. This conversion significantly decreases the memory required to load the base model into the GPU.

Quantized Low-Rank Adaptation (QLoRA) is a technique that synergizes these two approaches. It involves loading the frozen, pre-trained base model in a low-precision format (e.g., 4-bit NormalFloat) while keeping the small LoRA adapter weights in a higher precision for training [36]. This combination drastically reduces the overall memory consumption, making it possible to fine-tune very large models on a single GPU.

This efficiency, however, comes with a trade-off. While QLoRA substantially reduces memory requirements and allows for the use of larger models or larger batch sizes on the same hardware, it typically increases training due to the computational overhead of quantizing and de-quantizing weights during the forward and backward passes. This presents a critical decision for any research project: one can use standard LoRA on a smaller model for faster experimentation and iteration, or employ QLoRA to train a larger, potentially more capable model at the cost of longer training cycles. This choice is central to designing a feasible and effective experimental protocol.

2.6.3 Alternative PEFT methods

While a variety of PEFT methods have been developed, many are less suitable for the specific task of expanding Dutch dictionary definitions when compared to the more robust and widely adopted LoRA and QLoRA techniques. Methods such as prompt tuning, prefix-tuning and P-tuning, for instance, keep the base model's weights frozen and instead optimize a small set of prompt parameters that are prepended to the input sequence [37, 38]. Although highly parameter-efficient, these approaches can be less stable and more difficult to optimise than methods that directly adapt the model's weights. Achieving the nuanced and structured output required for high-quality lexicographical definitions through prompt manipulation alone presents a significant challenge, as the model's core generative knowledge remains unchanged.

Furthermore, other PEFT strategies are either too limited in scope or designed for different use cases. BitFit, which fine-tunes only the bias terms of the model, is extremely parameter-frugal but generally yields lower performance on complex generation tasks, as it fails to adjust the model's deeper representational knowledge [39]. Adapter-based methods, such as those proposed by Houlsby et al. (2019) and the cross-lingual framework MAD-X (Pfeiffer et al., 2020), insert small, trainable modules between the transformer layers [40, 41]. While effective, these methods introduce additional parameters and computational steps during inference, which can increase latency. Since the current research is focused on a monolingual generative task rather than multi-task or cross-lingual transfer, the architectural complexity of adapters offers little advantage over the more streamlined and performant approach of LoRA and QLoRA, which merge the trained weights back into the model for inference, adding no latency.

2.7 A hybrid evaluation framework

A robust and credible evaluation of expanded dictionary definitions requires a methodology that can assess quality from multiple perspectives. The task of definition expansion has two distinct success criteria: the generated definition must be semantically faithful to the core meaning of the concise input and it must be lexicographically appropriate in its style and structure. Automated metrics are highly effective at measuring the former by comparing generated text against a reference, but they are largely insensitive to the stylistic, structural and formal conventions that define high-quality lexicographical text. Therefore, a hybrid evaluation framework that combines automated metrics with a manual analysis is essential for a holistic assessment.

The quantitative component of the evaluation framework focusses on assessing semantic accuracy, drawing its metrics from the findings of the WMT24 Metrics Shared Task, a leading benchmark for evaluating text generation metrics [42]. A primary conclusion from this large-scale comparative study is the clear superiority of modern, fine-tuned neural metrics over traditional lexical overlap metrics like BLEU, particularly for evaluating the nuanced and fluent outputs of large language models. The results showed that neural metrics consistently rank higher and correlate better with human judgments across various domains and languages. However, BLEU and ROUGE are included for establishing a robust baseline. Their historical significance and continued widespread use in NLP provide a familiar frame of reference against which the performance of more sophisticated neural metrics can be contextualized and better understood. More importantly, their contrasting designs of precision-oriented versus recall-oriented offer a diagnostic lens through which to analyse the generative tendencies of the models.

While quantitative metrics are effective at measuring semantic accuracy, they are fundamentally incapable of assessing whether a text conforms to the stylistic and structural norms of a specific genre, such as a dictionary entry. A definition can be semantically perfect yet stylistically inappropriate, structurally flawed or otherwise unsuitable for a formal dictionary. To address this lexicographical appropriateness a systematic manual analysis grounded in the principles of dictionary criticism is required.

2.7.1 Foundational baseline with lexical overlap metrics

BLEU (Bilingual Evaluation Understudy) measures the quality of a generated text by calculating the precision of its n-grams (contiguous sequences of n words) relative to a set of reference texts. The core mechanism counts the proportion of n-grams in the hypothesis that also appear in the reference, adjusted by a brevity penalty to discourage outputs that are overly short [43]. In the context of definition expansion, BLEU serves as a proxy for precision and fluency. A high BLEU score suggests that the generated definition uses phrasing and lexical choices that align closely with those found in the reference ANW definitions, indicating an adherence to established lexicographical style. However, BLEU's limitations are significant, particularly for evaluating LLM outputs. It is unable to recognise synonyms or paraphrases as valid, penalising semantically correct but lexically diverse definitions [44].

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) was developed primarily for evaluating automatic summarization and operates on the principle of recall [45]. It measures the proportion of n-grams from the reference text that are successfully captured in the generated hypothesis. Variants such as ROUGE-L assess the overlap of in-sequence words without requiring them to be strictly adjacent, thus capturing structural similarity more flexibly. For the task of definition expansion, ROUGE functions as a measure of coverage and adequacy. A high ROUGE score indicates that the model has successfully incorporated the key informational components present in the reference full definition.

The joint analysis of BLEU and ROUGE provides a valuable diagnostic tool. A model that achieves a high BLEU score, but a low ROUGE score may be generating definitions that are precise and stylistically conventional but lack comprehensive detail; a conservative strategy. Conversely, a high ROUGE score paired with a low BLEU score suggests a model that captures all the necessary information but may do so in a verbose or stylistically novel manner. A model that scores well on both metrics demonstrates a balanced ability to be both comprehensive and stylistically appropriate.

2.7.2 State-of-the-art neural metrics for semantic assessment

To overcome the limitations of lexical overlap, the evaluation framework incorporates a suite of modern neural metrics. These models operate on the semantic meaning of the text rather than its surface form, providing a more accurate assessment of quality that aligns better with human judgment.

COMET-22 is a top-performing learned metric that predicts a quality score by creating and comparing representations of the source, hypothesis and reference texts using a powerful pre-trained encoder [46]. It is trained on large-scale human judgments of translation quality (direct assessments) and its scores have shown a consistently high correlation with human perception [42]. Within this framework, COMET-22 serves as the primary proxy for overall semantic quality. Its architecture is uniquely suited to this conditional generation task. By incorporating the source text (the short definition) into its calculation, COMET-22 does not merely assess the similarity between the generated full definition (hypothesis) and the original full definition (reference); it assesses this similarity conditioned on the input. This allows it to simultaneously evaluate the quality of the expansion and its faithfulness to the original concise meaning, making it an exceptionally well-suited metric for this task.

BERTScore complements COMET by providing a direct measure of semantic overlap that is robust to lexical variation. It computes the cosine similarity between the contextual embeddings of tokens in the hypothesis and the reference, generating precision, recall and F1 scores based on the optimal greedy matching of tokens [47]. The primary role of BERTScore is to fairly evaluate definitions that are semantically correct but use different wording, such as synonyms or paraphrases. Where BLEU and ROUGE would penalize a valid rephrasing, BERTScore can recognize the semantic equivalence, making it indispensable for a creative generation task where multiple correct expressions of a concept exist.

XCOMET represents a significant advancement in evaluation by providing a more transparent and explainable assessment. It is a learned metric trained not only on sentence-level scores but also on span-level human error annotations from the multidimensional quality metrics framework [48]. This dual training objective enables XCOMET to provide a high-quality overall score while also identifying and categorizing specific error spans within the generated text, such as inaccuracies or omissions. Its inclusion in this framework is motivated by its ability to bridge the gap between quantitative scores and qualitative analysis. By highlighting potential errors, XCOMET's output can directly inform and guide the manual qualitative review, making the hybrid evaluation process more efficient and integrated.

2.7.3 A systematic framework for manual analysis

Automated metrics, while effective for measuring semantic similarity, are fundamentally incapable of assessing whether a piece of text conforms to the stylistic and structural norms of a specific genre, such as a dictionary entry. To address lexicographical appropriateness a systematic manual error analysis is required. The methodology for this analysis is drawing upon the multi-level analysis framework proposed by Evert et al. (2024) [49]. Their work advocates for moving beyond simplistic rating scales and instead using a detailed evaluation grid that assesses specific quantitative and qualitative criteria for each component of a dictionary entry.

The proposed framework is adapted to an evaluation grid focused on the criteria most relevant to the task of definition expansion. This grid provides a concrete protocol for a human expert to annotate a sample of generated definitions, categorizing errors to provide qualitative insights that complement the automated scores. This structured approach ensures that the qualitative analysis is reproducible and systematic, rather than purely anecdotal. The grid (Table 1) is structured into three primary categories of analysis.

1. Semantic accuracy: This category assesses the factual and semantic correctness of the expansion, moving beyond similarity to judge truthfulness and accuracy.

- **Factual hallucination:** The expanded definition introduces new information that is factually incorrect or not implied by the concise input. This can be a critical failure point for generative models.
- **Omission:** A key semantic component from the input definition is missing in the expanded version, leading to an incomplete or misleading definition.
- **Sense drift:** The expansion subtly shifts the meaning towards a different sense of the same lemma, violating the core task constraint of sense-preservation.

2. Lexicographical quality: This category evaluates adherence to the formal and stylistic conventions of dictionary writing.

- **Stylistic appropriateness:** The language is too informal, conversational or otherwise stylistically inconsistent with a standard dictionary's objective tone. This is particularly relevant for evaluating models with conversational alignment, such as GEITje Ultra.

- **Structural correctness:** The definition fails to follow a clear and logical structure, such as the classic genus-differentia pattern (stating the general class of the concept, then its distinguishing features) where applicable.
- **Clarity and conciseness:** While the task is expansion, the output must still adhere to lexicographical principles of clarity, avoiding unjustified wordiness, redundancy or circularity.

3. Linguistic quality: This category assesses the fundamental grammatical and fluent properties of the generated text.

- **Fluency and grammar:** The generated text contains grammatical errors, awkward phrasing or is otherwise disfluent.
- **Naturalness:** The phrasing feels artificial, stilted or "machine-like," lacking the naturalness of human-written text.

Table 1: The evaluation grid for Dutch definition expansions.

Category	Criterion	Guiding Questions for Annotation
Semantic accuracy	Factual hallucination	Does the definition introduce information that is demonstrably false or not inferable from the source?
	Omission	Is a critical piece of information from the short definition absent from the full definition?
	Sense drift	Has the meaning shifted to a different sense of the lemma than the one specified by the input?
Lexicographical quality	Stylistic appropriateness	Is the tone objective, formal and encyclopedic? Is conversational or informal language used?
	Structural correctness	Does the definition follow a logical *genus-differentia* structure where appropriate? Is it well-organised?
	Clarity and conciseness	Is the definition easy to understand? Does it use more words than necessary or contain circular logic?
Linguistic quality	Fluency and grammar	Are there any grammatical, spelling or punctuation errors? Does the text flow naturally?
	Naturalness	Does the phrasing sound like it was written by a human expert or does it feel artificial and machine-like?

2.8 Conclusion

This literature review has surveyed the state-of-the-art across three critical domains for the task of Dutch dictionary definition expansion: foundation models, PEFT and evaluation methodologies. The synthesis of these domains provides a robust scholarly foundation for the proposed thesis and justifies its specific methodological choices.

The selection of mT5, Aya-101, Aya-23 and GEITje Ultra as candidate models is not arbitrary; it establishes a controlled experimental design to investigate key hypotheses. The comparison between the unaligned, encoder-decoder mT5 or the instruction aligned, decoder-only Aya models and the conversationally aligned, decoder-only GEITje allows for a direct test of the blank slate and domain mismatch hypothesises. The inclusion of Aya-101 versus Aya-23 provides a valuable data point on the "breadth vs. depth" trade-off in multilingual modelling, assessing whether a model with more focused multilingual training offers an advantage.

The decision to employ LoRA or QLoRA as the primary fine-tuning strategies is a pragmatic and powerful response to the computational constraints of academic research. These PEFT methods offer a path to adapting multi-billion parameter models on available hardware, with a clear and well-understood trade-off between memory efficiency and training speed.

Finally, the proposed hybrid evaluation framework is essential for a complete and nuanced assessment. By combining quantitative metrics for semantic accuracy with a structured qualitative analysis of lexicographical appropriateness, the thesis is positioned to deliver insights that a purely automated evaluation could not. This comprehensive approach ensures that the research will not only measure which model produces the most semantically similar text, but also which model produces the best dictionary definitions.

3 Methodology

This chapter details the methodological framework designed to answer the research questions. It provides a complete and reproducible account of the experimental procedures, beginning with the collection, cleaning and partitioning of the Dutch definition dataset. The chapter then outlines the two primary experimental paradigms: few-shot prompting and supervised fine-tuning both using QLoRA. For each paradigm, the prompt structures and training configurations are described in detail. Finally, it shows how the hybrid evaluation framework was used to assess the generated definitions.

3.1 Data

3.1.1 Data collection

The dataset for this study was compiled on the 28th of February 2025 from the Algemeen Nederlands Woordenboek (ANW), a comprehensive dictionary of modern Dutch provided by the Dutch Language Institute [50]. The data collection was performed programmatically by interacting with the public backend of the ANW, ensuring a systematic and reproducible process.

A foundational assumption for this research is that the dataset sourced from the ANW is authored entirely by human lexicographers, without the assistance of any Large Language Models. This premise is crucial to prevent a potential feedback loop of self-referential data, which would arise if a model were to be evaluated on definitions it or another model had a role in creating. While the ANW's public documentation does not specify the exact role of generative technology in its workflow, this assumption of human-only authorship is a necessary precondition for validating the experimental findings of this study.

The collection process began by downloading a complete list of all available lemmas from the ANW's JSON endpoint. This list, containing over 87,000 lemmas, served as the master index for the subsequent data extraction. The script then iterated through each lemma that was marked as having a definition. For each of these lemmas, a request was made to its corresponding URL to retrieve the full article data in XML format. This approach ensured that only entries with available definitions were processed, optimizing the efficiency of the data scraping.

Using the lxml library for robust parsing of the XML content, several key data fields were extracted for each definition sense associated with a lemma. The primary data points collected were the lemma itself, its part-of-speech (POS), the full, detailed definition and the concise, core-meaning definition. Additionally, unique identifiers for the lemma and the specific meaning were preserved to maintain the data's structural integrity. To manage the large scale of the extraction and ensure reliability, the process was implemented with a batching system and checkpointing, allowing the script to resume in case of interruption and methodically process all available definitions.

3.1.2 Dataset cleaning

The initial dataset consisted of 43,217 records. While comprehensive, this raw data presented a structural challenge for the task of definition expansion. A single record, representing one sense of a lemma, could be associated with multiple short definitions (concise meanings or synonyms) and several long definitions (full explanations). This many-to-many relationship is problematic when framing the task in a text-to-text paradigm. For the model to perform a consistent expansion task and for its output to be evaluated systematically, a one-to-one mapping from a single source definition to a single target definition was established as a methodological requirement. While it would be possible to concatenate multiple target definitions into a single output string, this would change the nature of the task from "definition expansion" to "generation of all possible definitions." Such an approach would create an inconsistent objective, where the model would have to generate a single definition for some inputs and multiple for others. This inconsistency significantly complicates automated evaluation and muddles the specific skill being tested. Therefore, a cleaning pipeline was developed to transform the raw data into a high-quality dataset of unambiguous one-to-one input-output pairs. The impact of the steps undertaken are summarized in Table 2.

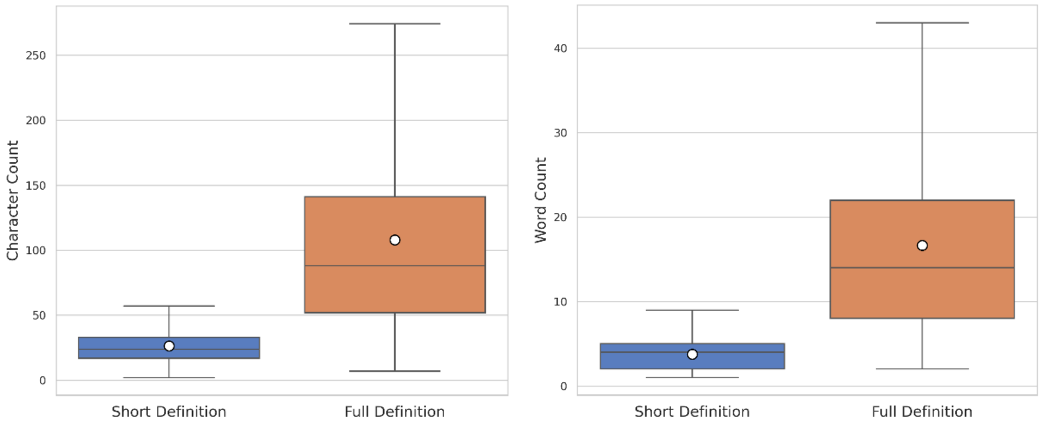
Table 2: The cleaning pipeline.

Cleaning Step	Rows Remaining	Rows Removed	Records Modified
Initial dataset	43,217	-	-
Removed missing values	38,005	5,212	-
Removed inconsistent POS tags	37,976	29	-
Removed identical definitions	36,374	1,602	12,440
Processed "ook: " prefixes	36,345	29	1,495
Normalization to 1-to-1 mapping	36,345	-	15,170
Applied final quality filters	34,824	1,521	-

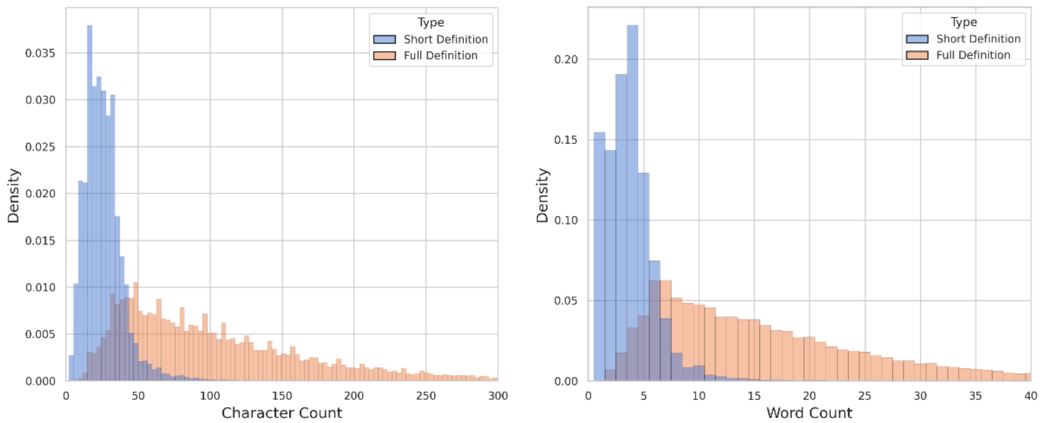
The cleaning process began with removing records with missing data or inconsistent POS tags to maintain dataset uniformity. Then any long definition parts that were exact copies of their short definition counterparts were removed as they did not suit the task of definition expansion. The focus then shifted to refining the definition content. Definitions prefixed with "ook:" (also:) were systematically processed; simple cases were salvaged by stripping the prefix, while more complex but infrequent variations were removed for efficiency as they constituted only a small portion of the affected definitions (364). The final transformation was normalizing the data to a one-to-one mapping by selecting the single best definition for each type. For short definitions, the one with the highest word count was chosen to prioritize descriptive phrases over synonyms, while for long definitions, the one with the greatest character length was selected to maximize the expansion task. Finally, a quality filter ensured that every record represented a valid expansion task where the full definition was longer than the short definition and consisted of more than a single word.

To prepare the data for the experiments, it was partitioned into a training set of 27,880 records (80%), a validation set of 3,494 records (10%), and a test set of 3,450 records (10%). A simple random 80-10-10 split would be inappropriate for both the fine-tuning and few-shot prompting methodologies. In a random split, different senses of the same lemma could appear across different sets. For fine-tuning, this would allow the model to learn lemma-specific information that could artificially inflate its performance on the test set. Similarly, for few-shot prompting, this could lead to a scenario where the prompt examples (drawn from the training set) contain the very same lemma that the model is being asked to define from the test set. This would not be a true test of generalization to novel words. Therefore, a strict lemma-level split was implemented. This method ensures that all records for a given lemma are confined to a single set (either train, validation or test), guaranteeing that the evaluation truly measures the models' ability to handle entirely unseen words.

This pipeline resulted in a final dataset of 34,824 records, representing 23,896 unique lemmas. The characteristics of this final dataset confirm its suitability for the task. As shown in the length distribution plots (Figures 3 and 4), there is a clear and significant difference between the length of the source (short) and target (full) definitions. The median short definition is 4 words long, while the median full definition is 14 words long (Figures 1 and 2), providing a distinct expansion signal for the models. The dataset is predominantly composed of nouns ("substantief"), which account for over 80% of the records (Figure 5). This provides a strong focus for the task by allowing the models to specialize on the linguistic patterns most common to defining this grammatical category. Furthermore, the dataset retains a rich representation of lexical semantics, with 5,818 lemmas being polysemous (having more than one sense), which ensures the models are exposed to a diverse and realistic sample of the Dutch lexicon (Figure 6).



Figures 1 and 2: Definition length boxplots measured respectively by character and word count (Outliers not shown for clarity).



Figures 3 and 4: Definition character and word length distribution plots.

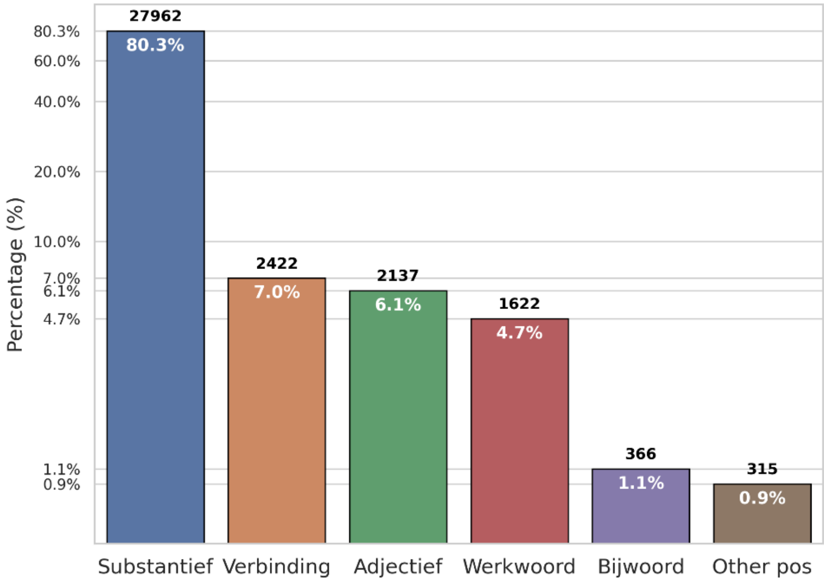


Figure 5: Part-of-speech distribution in the final dataset.

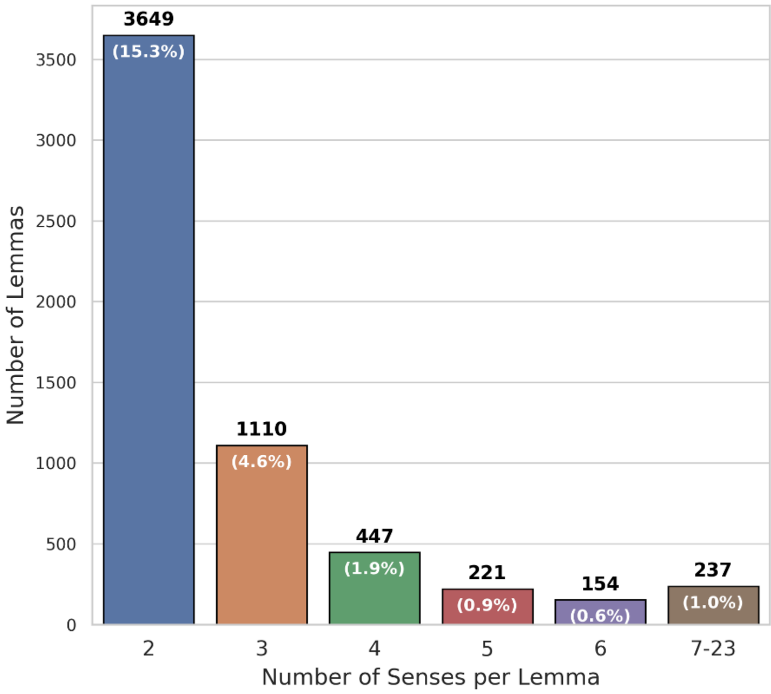


Figure 6: Distribution of senses for polysemous lemmas.

3.2 Experimental setup

The data preparation, cleaning and analysis were conducted within a standard Jupyter Lab notebook environment. For the computationally intensive tasks of model fine-tuning and inference, a high-performance cloud computing instance was utilised, equipped with a single NVIDIA A100 PCIe GPU (80GB VRAM). The software environment was based on a Docker image with PyTorch 2.2.0, Python 3.10 and CUDA 12.1.1; generative AI was also used as an assistant to generate and debug parts of the programming code. The primary challenge encountered was managing the significant memory requirements of the large models, which necessitated the universal adoption of QLoRA for all fine-tuning experiments to make training feasible on the hardware.

3.3 Few-shot prompting methodology

3.3.1 Few-shot example selection

To provide the language models with relevant in-context examples, a retrieval-augmented few-shot prompting strategy was employed. This approach proceeds in two distinct phases to select the most suitable demonstrations from the training set for any given test item.

The first phase involves semantic retrieval. To capture the conceptual meaning of each definition, every short definition in the training corpus was encoded into a vector representation using a multilingual SBERT model (paraphrase-multilingual-MiniLM-L12-v2), which is trained on over 50 languages including Dutch. These embeddings were then stored in a FAISS (Facebook AI Similarity Search) index, a library optimized for extremely fast similarity searching, to enable efficient retrieval. At inference time, the embedding of a test item's short definition is used to retrieve an initial pool of k=32 nearest neighbours based on cosine similarity. A pool of 32 was chosen to ensure a sufficiently diverse set of candidates for the subsequent re-ranking step. This strategy of selecting based on the short definition preserves the information boundary while effectively capturing topical proximity, a technique shown to outperform random sampling [51, 52].

In a second pass, these candidates are re-ranked using a hybrid scoring method. This method combines the initial semantic similarity with a lightweight lexical score computed on the lemmas using the token-set ratio, a language-agnostic form of fuzzy matching that compares shared words. Rewarding this morphological affinity helps to surface paradigm mates, such as “hartpatiënt” and “maagpatiënt”, which often share a common genus proxima (the general class of the concept) and require a similar structure for their differentia specifica (distinguishing features). This practice is adapted from translation memory prompting where high fuzzy matches yield consistent gains [53]. The final score is a convex combination of the two metrics: $s = \alpha \cdot cosine + (1 - \alpha) \cdot fuzzy$. The hyperparameter α was set to 0.8, a choice that heavily weights semantic similarity as the primary criterion for a good example, while using lexical similarity as a secondary signal to refine the ranking. The top three examples were selected for the final prompt, a number chosen as a pragmatic balance between providing sufficient context and minimizing computational overhead.

To mitigate the "lost in the middle" degradation effect, where models pay less attention to context in the middle of a long prompt, the selected examples were ordered with the most similar demonstration placed last, immediately preceding the test item [54]. The final prompt therefore provides the model with relevant, consistently formatted examples to guide its generation.

3.3.2 Prompt execution and output cleaning

The generated few-shot prompts were systematically applied to the test set for three selected models: GEITje Ultra, Aya-101 and Aya-23. These models were chosen based on the literature review to represent a range of modern architectures and training philosophies. The selection includes both an encoder-decoder model (Aya-101) and two decoder-only models (GEITje Ultra and Aya-23). They also embody different approaches to multilingualism: massive multilingualism (Aya-101), focused multilingualism (Aya-23) and language-specific continued pre-training (GEITje Ultra). The mT5 model was not included in this few-shot evaluation because, unlike the other models, it has not undergone instruction-tuning and is therefore not optimised to follow in-context examples without being fine-tuned first.

Each prompt provided the model with a clear task structure, as illustrated by the general template for “1,5 metermaatschappij” in Figure 7. The final formatting, including system prompts and model-specific chat tokens, was adapted to the requirements of each model.

Initial analysis of the model outputs revealed that the generations were not always limited to the expanded definition itself. Often, the models would replicate the formatting of the prompt, prepending the output with the target word, short definition and phrases like "Lange definitie:" or "Volledige definitie:". Since the goal is to evaluate the quality of the generated definition in isolation, a post-processing step was necessary. A script was implemented to clean these outputs by stripping away these predictable prefixes, isolating only the generated definition text.

```
Je bent een expert-lexicograaf die definities schrijft voor een Nederlands
woordenboek.

Voorbeelden:
Woord: anderhalvemetersamenleving
Korte definitie: samenleving waarin fysieke afstand nodig is
Lange definitie: samenleving waarin mensen die niet tot hetzelfde huishouden
behoren in de publieke ruimte een fysieke afstand van minimaal anderhalve
meter tot elkaar moeten bewaren om verspreiding van een virus te voorkomen

Woord: 1,5 metersamenleving
Korte definitie: samenleving waarin fysieke afstand nodig is
Lange definitie: samenleving waarin mensen die niet tot hetzelfde huishouden
behoren in de publieke ruimte een fysieke afstand van minimaal anderhalve
meter tot elkaar moeten bewaren om verspreiding van een virus te voorkomen

Woord: anderhalvemetermaatschappij
Korte definitie: maatschappij waarin fysieke afstand nodig is
Lange definitie: maatschappij waarin mensen die niet tot hetzelfde
huishouden behoren in de publieke ruimte een fysieke afstand van minimaal
anderhalve meter tot elkaar moeten bewaren om verspreiding van een virus te
voorkomen

Breid de volgende korte definitie voor het woord '1,5 metermaatschappij' uit
tot een volledige definitie: 'maatschappij waarin fysieke afstand nodig is'
```

Figure 7: Example of a few-shot prompt used for the base models.

3.4 Fine-tuning methodology

To adapt the foundation models to the specialized task of Dutch definition expansion, a supervised fine-tuning approach was employed. This method directly teaches the model the desired input-output behavior by training it on the prepared dataset. In addition to the three models subjected to few-shot prompting, mT5-xl was included in the selection for fine-tuning models as laid out by the literature review.

3.4.1 PEFT method selection

Given the significant size of these models, a full fine-tuning approach was computationally prohibitive on the available hardware. Therefore, a PEFT strategy was essential. Specifically, QLoRA was used across all models, with the paged_adamw_8bit optimizer. This technique involves loading the pre-trained base model in a quantized 4-bit format to drastically reduce its memory footprint, while only training small, LoRA matrices. The LoRA adapters were applied to the attention mechanism's query, key, value, and output layers (q, k, v, o) and the feed-forward layers (wi, wo) for the mT5-based models and to all linear layers (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj) for the decoder-only models. This approach makes it feasible to adapt multi-billion parameter models on a single GPU while preserving the vast knowledge of the original model.

3.4.2 Data formatting and training process

The partitioned dataset was loaded directly from the Hugging Face Hub. For each model, the training examples were formatted into a zero-shot instruction prompt (Figure 8). The structure of this prompt was adapted to the specific requirements of each model's architecture and pre-training. For the decoder-only, chat-oriented models (GEITje Ultra and Aya-23), the instruction was formatted using their respective chat templates, which included a system prompt to assign the model the role of an expert lexicographer. For the encoder-decoder models (mT5-xl and Aya-101), which are not chat-tuned, a more direct instruction without a system prompt was provided as the input.

The models were fine-tuned using the Hugging Face Trainer and SFT Trainer libraries. The training process was then closely monitored using Weights & Biases [55] to track metrics such as validation loss over time. To prevent overfitting and ensure the best-performing version of each model was selected, an early stopping callback was implemented with a patience of five evaluation steps. For Aya-101, Aya-23 and GEITje Ultra, this mechanism automatically concluded the training when the validation loss ceased to improve (after respectively 6900, 500 and 1300 steps). For mT5-xl, training was manually stopped after observing a prolonged plateau in validation loss, as further training did not yield improvements justifying the computational cost.

```
Je bent een expert-lexicograaf die definities schrijft voor een Nederlands
woordenboek.

Breid de volgende korte definitie voor het woord '1,5 metermaatschappij' uit
tot een volledige definitie: 'maatschappij waarin fysieke afstand nodig is'
```

Figure 8: Example of a zero-shot prompt used for the fine-tuned models.

3.4.3 Hyperparameters and configuration

After some initial experimentation, the key hyperparameters for the fine-tuning process were tailored to each model's architecture and size, as detailed in Table 3. The same models used for fine-tuning were used for few-shot prompting as well.

Table 3: Model hyperparameter configurations.

Aspect	**mT5-xl**	**Aya-101**	**Aya-23**	**GEITje Ultra**
Base Model	google/mt5-xl	CohereLabs/aya-101	CohereLabs/aya-expanse-8b	bramvanroy/geitje-7b-ultra
Architecture	Encoder-Decoder	Encoder-Decoder	Decoder-Only	Decoder-Only
Parameters	3.7B	13B	8B	7B
LoRA rank	32	16	32	32
LoRA alpha	64	32	64	64
Batch Size	2	2	4	4
Gradient Accumulation	8	8	4	4
Learning Rate	5e-5	2e-4	2e-4	2e-4
Training Stop	Manual (5500 steps)	Early (6900 steps)	Early (500 steps)	Early (1300 steps)

The hyperparameter configuration was established through a combination of best practices and adaptation. The settings for GEITje Ultra were based on the official recommendations provided on its Hugging Face model card, which were developed on a setup with four A100 GPUs, the same type of GPU used for this research [56]. These were adapted to the single-GPU setup by adjusting the batch size and gradient accumulation steps. These parameters proved to be a robust starting point and were also applied to the Aya-23 model. For the much larger Aya-101 model, the LoRA rank and alpha were reduced to further manage memory constraints. The encoder-decoder models (mT5 and Aya-101) required a higher gradient accumulation factor due to their larger memory footprint per training example compared to the decoder-only models. For mT5-xl specifically, a learning rate of 5e-5 was used, which is a widely adopted and effective baseline for fine-tuning T5-family models.

The significant difference in training duration, with the decoder-only models Aya-23 and GEITje stopping much earlier, can be attributed to their pre-training. As these models have already been heavily instruction- and chat-tuned, they likely adapt to the new, relatively simple instruction format of definition expansion much more quickly, reaching their peak performance on the validation set in fewer steps.

3.5 Evaluation framework

3.5.1 Automated metrics

Building upon the theoretical foundation established in the literature review a suite of automated metrics was implemented. The implementation relied on the Hugging

Face evaluate library, which provides a standardized framework for computing these scores.

The final selection of metrics provided a multi-faceted view of generation quality. Traditional n-gram overlap metrics, BLEU (for precision) and ROUGE (for recall), established a lexical baseline. To account for semantic equivalence beyond surface-level words, BERTScore was used to compare contextual embeddings. The primary metric for overall quality was COMET-22 (Unbabel/wmt22-comet-da), which is uniquely suited for this task as it evaluates the generated definition's quality conditioned on the short definition. Finally, XCOMET (Unbabel/XCOMET-XL) was included as a state-of-the-art quality estimation metric. While its error span categorization was explored, it was not included due to a technical challenge of implementing it. XCOMET's primary role here was to provide a high-quality holistic score for comparison since the manual analysis rendered automated error tagging less critical. Lastly, the model outputs were generated using beam search with a beam width of 4.

3.5.2 Qualitative analysis

The purpose of the manual analysis is not to score the entire test set, which would be infeasible, but to perform a deep evaluation on a strategically chosen sample of generated definitions. Given this practical constraint on the number of definitions that can be manually analyzed, a stratified sampling strategy was employed to maximize insight from a minimal number of examples. This approach is superior to random sampling, which would be inefficient and unlikely to surface the most informative cases for analysis.

The first stratum of the sampling strategy focused on high-performance analysis. To achieve this, the top-performing fine-tuned and few-shot models were identified based on their aggregate COMET-22 and XCOMET scores. From their outputs, the highest-scoring definitions that were not a perfect match with their reference definition were selected, so as to allow for meaningful analysis. The purpose of this stratum was to analyze the characteristics of high-quality outputs, thereby establishing a qualitative benchmark for what a successful definition expansion looks like and identifying the core strengths of the top-performing models.

The second stratum was designed to provide a direct comparative analysis. For each high-quality definition identified in the first stratum, the corresponding outputs for the exact same input (i.e., the same word and short definition) from the other top-performing model were also selected for review. This enabled a head-to-head comparison on the exact same lexical item, isolating the impact of the fine-tuning versus few-shot prompting approach and providing a clear view of their relative strengths and weaknesses.

The third stratum probed for structural consistency, specifically targeting the 'structural correctness' criterion from the evaluation grid (Table 1). This involved assessing whether models could learn and consistently generalize lexicographical patterns across related concepts. To this end, a semi-automated process was used to find pairs of lemmas from the test set sharing a clear genus proxima (general class). A script first identified candidate groups by searching for lemmas with common suffixes (e.g., ending in "-ologie"). From this generated list, lexicographically interesting pairs, such as “hematologie” (hematology) and “pneumologie” (pulmonology), were manually selected for the final analysis. While the selected pairs did not need to have the near-perfect aggregated scores required for the first stratum, they were still required to be high-quality outputs to ensure a meaningful comparison.

The analysis followed a rigorous protocol where each sampled definition was annotated using the evaluation grid (Table 1). These detailed annotations were then synthesized into a qualitative narrative that identified recurring error patterns, compared the typical behaviors of the two methods and provided illustrative examples.

3.6 Conclusion

This chapter has detailed the comprehensive methodological framework underpinning the empirical investigation of this thesis. A robust and reproducible pipeline was established, beginning with the programmatic collection and meticulous cleaning of the dataset from the Algemeen Nederlands Woordenboek. The implementation of a strict lemma-level data split is a critical feature of this design, ensuring that the model evaluations are not contaminated by data leakage and thereby providing a true measure of generalization to unseen lexical items. The subsequent sections laid out the two distinct experimental paradigms: retrieval-augmented few-shot prompting and PEFT with QLoRA. Which together facilitate a systematic comparison of different approaches to leveraging large language models for definition expansion.

A central component of this methodology is the hybrid evaluation protocol, which was designed to address the multifaceted nature of lexicographical quality. By combining a suite of state-of-the-art automated metrics, the semantic accuracy of the generated definitions can be quantitatively assessed. However, recognizing the inherent limitations of these metrics, the framework is augmented with a structured qualitative analysis. This manual review, guided by a detailed evaluation grid (Table 1), is essential for judging the stylistic appropriateness, structural correctness and overall lexicographical soundness of the model outputs.

In summary, the rigorous and multi-faceted methodology described provides a solid foundation for the experiments conducted in this study. Every step, from data preparation to the final evaluation, has been designed to ensure the validity of the findings and to enable a thorough investigation of the research questions.

4 Results

4.1 Introduction

This chapter presents the empirical findings from the experiments detailed in the previous chapter. The results are organised into two main sections. First, the quantitative results detail model performance on automated metrics, assessing lexical and semantic quality under both few-shot prompting and fine-tuning conditions. Second, the qualitative results examine the lexicographical quality of the generated definitions, identifying error patterns and notable behaviours based on the evaluation grid (Table 1).

4.2 Quantitative results

This section details the quantitative performance of the selected models on the test set comprising of 3,450 records. The results are organized by experimental condition, few-shot prompting and fine-tuning, to facilitate a clear comparison. The evaluation employs a hybrid set of metrics, detailed in the literature review. The primary metrics for semantic quality are COMET-22 and XCOMET, which are neural models trained to align with human judgments of quality. BERTScore-F1 complements these by measuring semantic overlap, making it robust to valid paraphrasing. For lexical similarity, the recall-oriented ROUGE-L is used to assess content coverage, while the precision-oriented BLEU score measures adherence to the specific phrasing of the reference definitions.

4.2.1 Performance of few-shot prompting models

The few-shot prompting experiments evaluated the in-context learning capabilities of three instruction-tuned models: Aya-101, Aya-23 and GEITje Ultra. The mT5-xl model was excluded from this experiment as its pre-training was purely unsupervised, making it unsuitable for following few-shot instructions without fine-tuning. The results are summarized in Table 4.

The results from the few-shot prompting experiments establish a clear performance hierarchy for the baseline condition. The Aya-101 model emerges as the undisputed leader, achieving the highest scores across all five metrics. Its notably high BLEU score of 0.2153 suggests its outputs often matched the reference phrasing very closely.

Conversely, the Aya-23 model performs exceptionally poorly in the few-shot paradigm. While its COMET-22 score remains competitive, its XCOMET, BERTScore, ROUGE-L and BLEU scores are much lower than those of the other models, indicating a significant failure to produce lexically and semantically relevant text under few-shot conditions. The GEITje Ultra model occupies a middle ground, with semantic scores comparable to the other models but with significantly lower lexical overlap scores than Aya-101. The performance of these models in the few-shot setting provides a crucial baseline against which the fine-tuned models can be compared.

Table 4: Quantitative evaluation of few-shot models.

Model	COMET-22	XCOMET	BERTScore-F1	ROUGE-L	BLEU
Aya-101	**0.6801**	**0.7650**	**0.7622**	**0.3920**	**0.2153**
GEITje Ultra	0.6703	0.6504	0.6955	0.2045	0.0536
Aya-23	0.6712	0.4878	0.6490	0.1004	0.0120

4.2.2 Performance of fine-tuned models

The fine-tuning experiments involved adapting four foundation models using the QLoRA methodology: mT5-xl, Aya-101, Aya-23 and GEITje Ultra. The performance of these models was assessed using the same suite of automated metrics. The consolidated results for the fine-tuned models are presented in Table 5.

Table 5: Quantitative evaluation of fine-tuned models.

Model	COMET-22	XCOMET	BERTScore-F1	ROUGE-L	BLEU
Aya-23	**0.6820**	**0.7559**	**0.7810**	0.3865	0.1336
GEITje Ultra	0.6661	0.7354	0.7795	**0.3871**	**0.1448**
Aya-101	0.6536	0.6298	0.7796	0.3780	0.1063
mT5-xl	0.6369	0.7100	0.7675	0.3494	0.1006

These results reveal a significant shift in the performance landscape compared to the few-shot baseline. The Aya-23 model, which performed poorly in the few-shot setting, achieves the highest scores on the primary semantic metrics, COMET-22 and XCOMET, as well as on BERTScore-F1 after fine-tuning. This indicates that its generated definitions were, on average, judged to be of the highest semantic quality and accuracy to the reference texts. The GEITje Ultra model also performs competitively, achieving the highest ROUGE-L and BLEU scores, suggesting strong content coverage and lexical precision, while its semantic scores are only slightly behind those of Aya-23. Conversely, Aya-101, the top performer in the few-shot paradigm, becomes a mid-tier performer after fine-tuning, lagging behind the smaller, decoder-only models. The mT5-xl model shows the lowest performance on the primary semantic quality metric, COMET-22. This divergence in model rankings between the fine-tuning and few-shot paradigms suggests that a model's aptitude for in-context learning is not necessarily predictive of its adaptability to supervised fine-tuning and vice versa.

4.2.3 Analysis of task-specific metrics

To provide a more granular view of model behavior, two additional metrics were employed. The first measures the rate of perfect replication of the reference definition, serving as another measure of accuracy. The second quantifies the models' adherence to the implicit task format in the few-shot setting.

Accuracy via perfect matches

A "perfect match" is defined as an instance where a model's, cleaned, generated definition is identical to the reference full definition. This metric provides an unambiguous measure of a model's ability to produce the exact target output. A "normalized match" variant was also calculated, which ignores capitalization of the first letter and the presence of a single trailing period, offering a slightly more lenient assessment of accuracy. The results for all models across both conditions are presented in Table 6.

Table 6: Comparison of perfect and normalized matches across all models out of 3,450 definition expansions.

Model	Exact Match	Exact Match (%)	Normalized Match	Normalized Match (%)
Aya-101 FS	**322**	**9.33%**	**341**	**9.88%**
GEITje FT	277	8.03%	277	8.03%
Aya-23 FT	258	7.48%	258	7.48%
Aya-101 FT	247	7.16%	247	7.16%
mT5-xl FT	166	4.81%	166	4.81%
Geitje FS	2	0.06%	7	0.20%
Aya-23 FS	0	0.00%	0	0.00%

The data reveals that the few-shot Aya-101 model achieved the highest rate of both exact and normalized matches, successfully replicating the reference definition in nearly 10% of cases. This aligns with its high BLEU score and suggests a strong capability for lexical mimicry when prompted with in-context examples. Among the fine-tuned models, GEITje Ultra and Aya-23 performed the best, generating perfect matches over 8% and 7% of the time, respectively. The few-shot versions of GEITje Ultra and Aya-23 performed extremely poorly on this metric, with Aya-23 failing to produce a single perfect match. This highlights a clear performance gap, where fine-tuning consistently enabled models to generate precise, target-like outputs far more effectively than few-shot prompting for all models except Aya-101.

Task adherence in few-shot prompting

During the few-shot experiments, it was observed that models often replicated the formatting of the prompt examples, prepending their outputs with phrases like "Lange definitie:". A post-processing step was required to strip this extra text. The frequency of this behavior serves as a behavioral metric, acting as a proxy for how well each model adhered to the implicit instruction of generating only the definition text itself. Table 7 summarizes these cleaning statistics.

Table 7: Cleaning statistics for few-shot models of 3,450 definition expansions.

Model	Unchanged Outputs	Adapted Outputs	Percentage Adapted (%)
Aya-23	246	3204	92.87%
Aya-101	1801	1649	47.80%
GEITje	**2766**	**684**	**19.83%**

The results show a dramatic difference in model behavior. The Aya-23 model required cleaning in over 92% of its outputs, indicating it almost always mimicked the full format of the prompt examples rather than isolating the core task. In contrast, the GEITje Ultra model was far more direct, requiring cleaning in less than 20% of cases, suggesting it better understood the underlying instruction to simply provide the expanded definition. Aya-101 fell in the middle, requiring adaptation for nearly half of its outputs. This quantitative measure of task adherence reveals fundamental differences in the instruction-following styles of the models, with GEITje Ultra exhibiting a more direct response style and Aya-23 a more imitative one.

4.3 Qualitative results

While quantitative metrics provide a valuable overview of semantic and lexical similarity, they cannot fully capture the multifaceted nature of lexicographical quality. To gain a deeper understanding of the models' capabilities, a systematic qualitative analysis was performed on a stratified sample of generated definitions. This analysis, guided by the evaluation framework from the literature review, synthesizes the detailed manual evaluations. The full, case-by-case evaluations that form the basis of this synthesis can be found in appendix A.

4.3.1 Comparative synthesis of top-performing models

Semantic accuracy

The analysis confirmed that Aya-101 (Few-shot) was capable of producing outputs of exceptional quality, including perfect matches for lemmas like *ecoroman* and *lichtgroen*. However, its performance was inconsistent. When it failed, it often failed completely, resulting in severe semantic omission, where the output was a non-expansion or a direct copy of the input (e.g., for *benoemingstermijn*, the output was identical to the short definition). In the most severe cases, which were few, it exhibited major sense drift and factual hallucination, as seen with *kleutertuin* where it provided a completely incorrect definition for a different concept.

In contrast, Aya-23 (Fine-tuned) demonstrated far greater reliability. While it produced fewer perfect matches than the few-shot Aya-101, it consistently performed the expansion task. Its errors, few as well, were typically more subtle. For example, with *groepsspreekuur*, it produced a plausible but slightly incorrect definition, a form of minor sense drift. For *ecoroman*, it captured the core meaning but omitted a secondary detail ("andere milieuvraagstukken"), an instance of minor semantic omission. It was not observed to produce complete non-expansions or catastrophic sense drifts in the sampled data.

Lexicographical quality

Both models generally produced fluent, grammatically correct Dutch. However, the few-shot model was more prone to stylistic errors. For inputs like *trouwplanner*, it failed to expand the definition and instead reframed the input as a complete sentence ("Een trouwplanner is iemand die bruiloften organiseert."), which is stylistically inappropriate for a dictionary entry. It also occasionally included the lemma in the definition itself (e.g., "Aspergerobot: machine om asperges te oogsten"), a significant formatting error. The fine-tuned Aya-23 was more consistent in adhering to the expected lexicographical format, though it sometimes produced circular definitions, as seen with *lifestylecoach*, where it defined the term using "leefstijl" (lifestyle) without providing specific details.

A rare but notable error type for the few-shot model was minor factual hallucination, where it added plausible but unverified information not present in the reference. For *Wageningse* ("woman from Wageningen"), it added "uit de gemeente Wageningen" ("from the municipality of Wageningen") and for *dorpswinkelier* ("village shopkeeper"), it added "of als vrijwilliger" ("or as a volunteer"). This tendency to add information was not observed in the fine-tuned model's outputs.

4.3.2 Structural Consistency in Genus Proxima Groups

To assess the models' ability to generalize lexicographical patterns, their performance was evaluated on groups of related terms sharing a common genus proxima (the general class of the concept). The analysis focused on two groups: nouns ending in -*liefhebber* (enthusiast) and medical specialisms ending in -*ologie* (the study of).

The results showed that the fine-tuned model, Aya-23, demonstrated superior and more consistent structural generalization. For the -*liefhebber* group, it correctly identified the core expansion pattern (e.g., expanding "liefhebber van X" to "iemand die veel van [activiteit met X] houdt") for all tested lemmas, including *operaliefhebber*, *sportliefhebber* and *theaterliefhebber*. While the expansions were sometimes incomplete (omitting "spelen" for *theaterliefhebber*) or slightly drifted ("zingen" for *operaliefhebber*), the model consistently applied the correct underlying structure.

The performance of the few-shot model, Aya-101, was directly correlated with the quality of the examples retrieved for its prompt. When provided with a perfect structural analogue in the prompt (e.g., receiving *toneelliefhebber* as an example for theaterliefhebber), it produced an excellent, well-structured definition. However, when the retrieved examples were only thematically related but structurally dissimilar (e.g., receiving *taalkundige* as an example for *taalliefhebber*), the model failed to generalize and defaulted to a minimal, unexpanded output.

This pattern was even more pronounced in the -*ologie* group. Both models performed excellently on *nefrologie* (nephrology). The few-shot model did so because it was provided with strong structural examples in the prompt. For *hematologie* (hematology), the few-shot model produced a near-perfect definition after being prompted with *reumatologie*. However, for *pneumologie* (pulmonology), where the retrieved prompt examples were thematically relevant (*longstayer*, *pneumonie*) but lacked the correct "-logie" definition structure, the few-shot model failed completely, producing a non-expansion. The fine-tuned model, in contrast, successfully applied the correct expansion pattern to *pneumologie*, demonstrating its ability to generalize without reliance on in-context examples. This indicates that while the few-shot approach can succeed when guided by highly relevant examples, the fine-tuned approach has more robustly internalized the structural patterns required for the task.

4.4 Conclusion

The results presented in this chapter provide a multi-faceted view of the models' performance. The quantitative evaluation established a clear hierarchy within each experimental paradigm. For fine-tuning, Aya-23 emerged as the top-performing model based on semantic quality metrics, closely followed by GEITje Ultra. For few-shot prompting, Aya-101 was the undisputed leader, outperforming the other models on every metric and achieving the highest rate of perfect matches of any model in the study.

The qualitative analysis complemented these findings by revealing the distinct behavioral profiles of the top models. The fine-tuned Aya-23 was characterized by its consistency and reliability, consistently producing structurally sound expansions, with its errors being typically minor and related to refinement. In contrast, the few-shot Aya-101 exhibited a hit or miss pattern: it was capable of producing perfect outputs but was also prone to complete task failure, with its success being highly dependent on the quality of the examples provided in its prompt.

5 Discussion

5.1 Introduction

This chapter synthesizes and interprets the quantitative and qualitative results presented in the previous chapter. The primary objective of this research was to assess the viability of modern transformer-based models for the specialized lexicographical task of expanding concise Dutch dictionary definitions into comprehensive, formally structured entries. The analysis that follows seeks to provide comprehensive answers to the three core research questions by dissecting the performance of the selected models under both few-shot prompting and fine-tuning paradigms.

The discussion is structured to build a cohesive argument. First, it will deconstruct the performance of the few-shot prompting approach, arguing that it is fundamentally ill-suited for this high-precision domain due to issues of task adherence and stylistic mismatch. Second, it will contrast this with the transformative impact of supervised fine-tuning, demonstrating how task-specific training instils the necessary domain discipline. The chapter will then introduce two concise sections: one justifying the hybrid evaluation framework and another critically examining a key study limitation revealed by an anomalous model performance. Finally, the chapter will synthesize these findings to discuss the broader implications for the field of computational lexicography and the evolving role of the human expert.

5.2 The inefficacy of few-shot prompting

The first research question sought to establish a baseline by evaluating how effectively modern LLMs can generate lexicographically sound definitions using only few-shot prompting. The quantitative results, presented in Table 4, initially suggest a clear performance hierarchy, with Aya-101 achieving the highest scores across all metrics, including a XCOMET score of 0.7650 and a remarkable BLEU score of 0.2153. However, a deeper qualitative analysis reveals that while the method is generally unreliable, the performance of Aya-101 is a notable exception. Its success, though inconsistent, suggests that with the right model and carefully selected examples, few-shot prompting can be a viable strategy. The failures of the other models are rooted in poor task adherence and a stylistic mismatch stemming from their pre-training, but Aya-101's pre-training appears to be a better fit for this task.

A primary failure of the few-shot paradigm was the models' inability to adhere to the implicit task format, a direct consequence of their pre-training. Both GEITje and the Aya models have undergone extensive alignment for conversational or general instruction-following interaction, respectively. This creates a strong stylistic prior that conflicts with the formal, non-conversational nature of lexicographical text. The evidence for this mismatch is starkly quantified in Table 7, which details the output cleaning statistics. The Aya-23 model required its output to be adapted in over 92% of cases, indicating that it almost always mimicked the full structure of the prompt examples rather than generating only the definition text itself. This behavior suggests its instruction-tuning has optimized it for meticulous format replication, a trait of a helpful assistant, but one that fails the implicit goal of this task. Conversely, GEITje Ultra required cleaning in less than 20% of cases, suggesting its alignment has produced a model more adept at inferring the user's underlying intent (to provide the definition directly) rather than simply copying the prompt's structure. This divergence shows that instruction-aligned pretraining is not a universally applicable process and can result in different behavioral failure modes, making few-shot prompting an unreliable method for enforcing a specific output format.

Aya-101, however, presents a more complex and promising case. Its high quantitative scores (Table 4) and the highest rate of perfect matches in the entire study at 9.33% (Table 6) paint a picture of remarkable capability. The qualitative analysis reveals a "hit-or-miss" profile characterized by extreme inconsistency. This suggests that the model is not suffering from a fundamental inability to perform the task, but rather from the selection of the few-shot examples. When the examples align well with the target lemma, Aya-101 produces excellent, lexicographically sound definitions. For instance, for the lemma *ecoroman*, it generated a perfect definition. However, it failed catastrophically on other lemmas, such as the major sense drift and factual hallucination for *kleutertuin*. This binary performance pattern is not indicative of a model struggling with its core capabilities, but rather a model that is highly sensitive to the provided examples. The fact that the other two few-shot models performed poorly with the same examples further suggests that Aya-101's pre-training is uniquely suited for this task. It is a formidable contender and its inconsistency is likely a solvable problem, pointing to the critical importance of few-shot example selection in this methodology.

5.3 The transformative impact of fine-tuning

The second research question asked how the performance of PEFT LLMs compares to the few-shot prompting baseline. The evidence demonstrates that fine-tuning is not merely an incremental improvement but a necessary and transformative process. It reshapes a model's behavior, overwriting its general-purpose priors and instilling the domain-specific discipline required for reliable lexicographical generation.

The quantitative results show a dramatic turnaround, best exemplified by the Aya-23 model. In the few-shot setting, it was the worst performer on lexical overlap metrics. After fine-tuning it became the top performer on the key semantic quality metrics, with its XCOMET score rising from 0.4878 to 0.7559 and its BERTScore-F1 jumping from 0.6490 to 0.7810 (Table 4 and Table **5**). This reversal proves the power of fine-tuning to adapt a model to a specialized task.

The true impact of fine-tuning is most evident in the models' ability to generalize lexicographical patterns, moving beyond simple pattern mimicry like with few-shot prompting. The genus proxima analysis (appendix A.2) provides compelling evidence of this shift. The fine-tuned Aya-23 model consistently identified and applied the correct underlying expansion structure for related terms. For the *-liefhebber* (enthusiast) group, it correctly expanded terms like *operaliefhebber* and *sportliefhebber* by capturing the core pattern of "iemand die veel van [activity] houdt." Similarly, for the *-ologie* (study of) group, it successfully generated structurally sound definitions for *nefrologie* and *pneumologie* by applying the learned pattern of "medisch specialisme dat zich bezighoudt met...". This demonstrates that the model has internalized a generalizable rule.

In stark contrast, the few-shot Aya-101 model's success was entirely contingent on the quality of the examples retrieved for its prompt. It produced an excellent definition for *theaterliefhebber* only because it was prompted with the structurally identical *toneelliefhebber*, but it failed on *taalliefhebber* when the retrieved examples were only thematically related. This reveals that few-shot learning operates closer to concrete pattern mimicry than true generalization. Fine-tuning encourages the learning of these robust structural patterns, leading to far more reliable and consistent behavior.

Finally, the results of the fine-tuning experiment provide empirical support for the "depth vs. breadth" strategy in multilingual model design. The 8-billion parameter Aya-23, which focuses model capacity on 23 languages, clearly outperformed the larger 13-billion parameter Aya-101, which covers 101 languages, on this Dutch-specific task. Since the fine-tuning process was identical, this performance difference stems from the base models' pre-training. This offers a guideline for model selection: for high-performance tasks in a non-English language, a model with a more focused multilingual pre-training strategy is preferable to a "broad" but shallow one, even if the latter is larger.

5.4 Indispensability of a hybrid evaluation framework

The third research question concerned the development of an effective evaluation framework. The findings of this study validate that the proposed hybrid protocol, combining automated metrics with qualitative analysis, was not just effective but essential. A purely quantitative approach would have led to fundamentally flawed conclusions, proving that qualitative analysis is indispensable for understanding true model capability in nuanced generative tasks.

The most compelling evidence for this is the case of the few-shot Aya-101 model. Based solely on the automated metrics in Table 4 andTable **6**, it would be declared the best-performing model in the study. However, the qualitative analysis revealed its "hit-or-miss" profile: a mix of perfect outputs and complete failures, rendering it unreliable for any practical application when not given the appropriate few-shot examples. This discrepancy between the quantitative scores and the qualitative reality confirms that human-in-the-loop evaluation is a necessary safeguard.

Furthermore, the structured qualitative analysis was necessary to identify specific categories of lexicographical errors that automated metrics are blind to. For example, the fine-tuned Aya-23 model produced a circular definition for *lifestylecoach*, defining the term with "leefstijl" (lifestyle) without providing specific details. An automated metric like BERTScore assigns this a high score (0.7804) for semantic overlap, but a human expert correctly identifies it as an uninformative definition. Similarly, for *dorpswinkelier*, the few-shot Aya-101 model exhibited minor factual hallucination by adding the plausible but unverified detail "of als vrijwilliger". This fluent, semantically related addition would not be penalized by automated metrics but constitutes a factual error from a lexicographical standpoint.

Finally, employing a suite of metrics provides a richer, more diagnostic profile of model behavior than any single metric could. The divergence in scores between the top-performing fine-tuned models in Table 5 illustrates this point. The Aya-23 model achieved the highest score on the primary semantic quality metric, COMET-22 (0.6820), while the GEITje Ultra model scored highest on the lexical overlap metrics ROUGE-L (0.3871) and BLEU (0.1448). This is not a contradiction but a key finding. It suggests that Aya-23 is more adept at generating definitions that are judged as semantically superior, even if they deviate lexically from the reference, whereas GEITje is better at the lexical mimicry of the training data. A single metric would have obscured this crucial behavioral difference. Therefore, the hybrid framework provides a more complete, diagnostic profile of each model and thus confirming its indispensability for a nuanced evaluation.

5.5 Synthesis and model selection

This study's model selection was designed to test several key dichotomies: encoder-decoder vs. decoder-only architectures, unsupervised vs. aligned pre-training and broad vs. focused multilingualism. The results provide clear takeaways. Fine-tuning proved superior to few-shot prompting across the board. The focused multilingual strategy of Aya-23 yielded better results for Dutch than the broad strategy of Aya-101. And while a "blank slate" model like mT5 avoids conversational bias, a stronger, instruction-tuned base model like Aya-23 ultimately provided a better foundation for fine-tuning. Synthesizing these findings allows for a definitive selection of the best overall approach for this task. Guided by the principle that for any practical application, consistent reliability is far more valuable than sporadic excellence the overall winner is the fine-tuned Aya-23 model.

The justification for this choice is rooted in the qualitative evidence, which highlights the critical difference between reliability and raw scoring power. The fine-tuned Aya-23 model is, above all, consistent. Its outputs are structurally sound, its adherence to lexicographical style is strong and its errors, when they occur, are typically minor and predictable, such as the slight omission for *ecoroman*. This makes it a trustworthy tool that can genuinely assist a lexicographer in their workflow.

In contrast, the few-shot Aya-101 model, despite its high scores, is fundamentally untrustworthy. The genus proxima analysis demonstrated that its success was attributable to pattern mimicry and the selection of the few-shot examples. A lexicographer using this model would have to treat outputs with more suspicion, checking for non-expansions, stylistic errors or hallucinations like the one seen for *kleutertuin*. This higher need for verification would completely negate the efficiency benefits of automation.

The choice, therefore, comes down to practical utility. The consistent and more reliable performance of the fine-tuned Aya-23 model makes it a superior and genuinely useful tool for the task of automated definition expansion when the evaluation metrics score high. It produces high-quality first drafts that a human expert can trust and refine, effectively shifting their role from author to editor and validator. The "hit-or-miss" of the few-shot Aya-101, with its unpredictable mix of perfection and failure, offers less such practical advantage.

5.6 Conclusion

This chapter has interpreted the empirical results of the study to provide comprehensive answers to the guiding research questions. The analysis confirms that while modern LLMs possess powerful generative capabilities, their effective application to a specialized, high-precision domain like lexicography is not straightforward.

The findings can be summarized as follows. First, few-shot prompting is a less effective strategy for this task. The domain mismatch between the models' conversational or general-purpose alignment and the formal requirement of lexicography leads to poor task adherence and stylistic incongruity. Second, supervised fine-tuning is a necessary and transformative step. It successfully instills the required domain-specific discipline, enabling models to produce reliable, structurally correct and stylistically appropriate definitions. Third, a hybrid evaluation framework that combines automated metrics with structured qualitative analysis is indispensable. It provides a safeguard against misleading scores and is essential for diagnosing nuanced errors that automated metrics cannot detect.

The central contribution of this work is the clear demonstration that for specialized, formal domains, task-specific fine-tuning is the key to unlocking the practical potential of LLMs. It is the most effective method for overcoming the stylistic priors of general-purpose models and achieving reliable domain-appropriate output. For the practice of lexicography, these findings are highly significant. Fine-tuned models like Aya-23 can serve as powerful and practical "lexicographer's assistants," capable of generating high-quality first drafts that can significantly accelerate the workflow and improve the consistency of large-scale dictionary projects like the ANW. However, this optimism must be tempered: the persistence of subtle errors, even in the best-performing fine-tuned model underscores that LLM generated content still requires critical review.

6 Conclusion

This final chapter synthesizes the findings of the research, offers a critical reflection on the methodology and its implications, and outlines promising directions for future research. The central objective of this thesis was to systematically evaluate the efficacy of modern LLMs for the specialized lexicographical task of expanding concise Dutch dictionary definitions into formally structured entries. Through a series of controlled experiments, this work has generated clear answers to its guiding research questions, contributing to a deeper understanding of the practical application of LLMs in a high-precision domain.

6.1 Summary of findings and contributions

The first research question sought to determine the effectiveness of few-shot prompting as a low-resource strategy for definition expansion. The findings indicate that this approach is largely unreliable for the task. This inefficacy stems from a fundamental domain mismatch between the models' pre-training, which typically optimizes for conversational or general instruction-following, and the formal, non-conversational requirements of lexicography. Evidence for this mismatch was apparent in the models' poor task adherence; the instruction pretrained Aya-23 model, for instance, mimicked the prompt's formatting in over 92% of its outputs, demonstrating a tendency to replicate structure rather than perform the core task. While the few-shot Aya-101 model achieved the highest quantitative scores in the whole study, its performance profile was characterized by a "hit-or-miss" inconsistency, ranging from perfect outputs to complete failures. This unreliability renders it unsuitable for any practical, real-world application where consistency is paramount.

The second research question compared the few-shot baseline against the performance of PEFT LLMs. The evidence demonstrates that supervised fine-tuning has a transformative impact, proving to be a superior and necessary approach for this domain. This is best exemplified by the dramatic performance reversal of the Aya-23 model, which evolved from the worst-performing few-shot model on lexical metrics to the top-performing fine-tuned model on key semantic quality metrics. A crucial finding is that fine-tuning enables models to move beyond the simple pattern mimicry characteristic of few-shot learning towards a more robust internalization of lexicographical rules. The success of the few-shot model was shown to be entirely contingent on being prompted with a near-perfect structural analogue. In contrast, fine-tuning, which involves updating model weights across thousands of diverse examples, compels the model to learn a generalizable, compressed representation of the task's underlying structural patterns. This capability was confirmed in the analysis of genus proxima groups, where fine-tuned models consistently generalized expansion patterns to related terms, a feat the few-shot model could only achieve when provided with ideal examples.

The third research question concerned the development of an effective evaluation framework. This study validates the indispensability of the proposed hybrid protocol, which combines automated metrics with structured qualitative analysis. Relying solely on quantitative scores would have led to the fundamentally flawed conclusion that the few-shot Aya-101 was the best overall model. The qualitative analysis was essential in revealing its underlying unreliability. Furthermore, this manual review was necessary to identify nuanced lexicographical errors that automated metrics are blind to, such as the generation of circular definitions or minor factual hallucinations, thereby providing a more complete and accurate assessment of true model capability.

6.2 Critical evaluation and implications

While this thesis provides clear results, a critical reflection on its methodology reveals important limitations. The few-shot prompting strategy, though sophisticated in its design, contained a significant methodological shortcoming. The qualitative analysis revealed that few-shot performance is extremely sensitive to the structural quality of the retrieved examples. The hyperparameter α, which controls the balance between semantic relevance and structural similarity in the example selection process, was therefore a critical variable. However, its value was set to 0.8 based on intuition and was never systematically tested against other values. It is plausible that a different balance, perhaps one that placed greater weight on lexical and structural similarity, could have improved the consistency of the few-shot results by retrieving better structural analogues for the model to mimic. This represents a missed opportunity for optimization and a key lesson in methodological rigor. Similarly, the number of few-shot examples was set at three without examining the impact of increasing the examples shown in the prompt.

A more profound challenge concerns the source of the dataset itself. The anomalous and inconsistent performance of the few-shot Aya-101 model compared to the other few-shot experiments invites a critical hypothesis: that its high scores could be an artifact of data contamination. It is conceivable that a portion of the Algemeen Nederlands Woordenboek (ANW) dataset, which formed the basis of this study, was created with the assistance of an LLM, potentially even a precursor to or version of the Aya-101 model. As noted in the data collection, the interpretation of the results assumes the dataset was not created with help of any LLM, but this is an unverified assumption. If this hypothesis of data pollution were true, the implications would be significant. The model's high performance would not be evidence of genuine lexicographical generalization through in-context learning but rather of its ability to recognize and replicate its own machine-generated patterns present in the ground-truth data. Its "hit-or-miss" behaviour could then be explained as the difference between encountering a familiar, LLM generated pattern (a "hit") and a novel, genuinely human-written definition for which its retrieved examples were poor (a "miss"). This possibility not only alters the interpretation of this study's results as it would put forth the fine-tune approach as the even more definite superior method. But it also highlights a fundamental challenge for the entire NLP field: the problem of benchmark integrity in an era where models are increasingly trained on the output of other models, creating a potential feedback loop of self-referential data.

6.3 Limitations and future work

The limitations and critical evaluations discussed above provide a clear roadmap for future research. The primary recommendation is to address the uncertainty surrounding the sourced dataset. Future work in this area must prioritize the diversity of lexicographical data to minimize the impact of potential LLM use to generate parts of the data to ensure the validity of the evaluation. To enhance model robustness and generalization, training corpora should be diversified by incorporating definitions from a wider range of dictionaries, exposing models to more varied lexicographical styles and counteracting the potential homogenizing effect of a single, potentially LLM assisted source.

Furthermore, limitations within the evaluation protocol itself present clear avenues for future improvement. A notable technical challenge was the inability to fully operationalize the error span categorization feature of the XCOMET metric. Future work should prioritize the implementation of this feature, as it would provide a more systematic guide for the manual analysis and more effectively bridge the quantitative and qualitative assessments. Relatedly, the qualitative analysis in this study was intentionally focused on high-performing outputs to establish a benchmark for success. A more comprehensive future study, aided by automated error tagging, should also investigate common failure modes in lower-scoring definitions to build a complete taxonomy of model errors.

Acknowledging the rapid evolution of the field, a natural next step is to replicate this study with newer, more powerful and Dutch-centric models, such as the forthcoming GPT-NL model. This would help determine if increased scale and more advanced, language-specific pre-training can mitigate some of the limitations observed here. In parallel, to further probe the potential of low-data methods, future research should investigate more sophisticated prompting techniques, such as chain-of-thought prompting or the inclusion of explicit stylistic rules within the prompt itself. Critically, such work must also include a systematic exploration of key methodological hyperparameters, such as the α value in the example selection process, which was identified as a limitation in this study.

In conclusion, despite the identified challenges, this research strongly supports the potential for fine-tuned LLMs to function as powerful "lexicographer's assistants." The findings clearly indicate that for specialized, formal domains, task-specific fine-tuning is the key to unlocking the practical potential of these models by overcoming the stylistic priors of their general-purpose pre-training. The path forward lies in developing synergistic human-in-the-loop systems, where the human expert's role evolves from that of a primary author to that of a critical validator and refiner.

6.4 Concluding note

In summary, this thesis investigated the potential of transformer-based models to automate the expansion of concise Dutch dictionary definitions, empirically comparing few-shot prompting against parameter-efficient fine-tuning. The results demonstrated that while few-shot prompting can achieve high scores, it is fundamentally unreliable for this high-precision task. In contrast, fine-tuning proved to be a transformative approach, enabling models to robustly learn and apply the necessary lexicographical patterns with high consistency. Despite the limitations discussed, the findings provide a viable methodology and a valuable resource for future work in developing human-in-the-loop systems and assistive tools for lexicography. The complete codebase, including scripts for data processing, model training and evaluation, is accessible in the project's GitHub repository[1]. The dataset used for this research can be found on the Hugging Face Hub[2].

[1] Code is released under: https://github.com/RobbeDoesPy/Dutch-Dictionary-Definitions-Expansion
[2] Dataset splits are released under: https://huggingface.co/datasets/RobbedoesHF/dutch-definitions

APPENDICES

A Qualitative analysis

A.1 Top-performing definitions

This appendix provides a detailed comparative analysis of the top 20 unique definitions generated by the fine-tuned model (Aya-23) and the few-shot model (Aya-101). Each sample was evaluated against a reference definition to assess its semantic accuracy, stylistic appropriateness and overall quality. Table 1: The evaluation grid for Dutch definition expansions. was used to accomplish this.

Sample 1: *liefhebber* (Triggered by high score from Aya-101)

- **Short definition:** iemand die veel van iets of iemand houdt
- **Long definition:** iemand die iets of iemand heel leuk of mooi vindt

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
    - **Prediction:** iemand die veel van iets of iemand houdt
    - **Scores:** COMET-22 = 0.8541, XCOMET = 0.9623
    - **Evaluation: Semantic Omission.** The prediction is a direct copy of the short definition, failing to perform the expansion task. It omits the nuance of finding something "leuk of mooi" (nice or beautiful) present in the reference. However, the long definition it self doesn’t differ too much from the short definition, making this expansion hard.
- **Aya-101 (Few-Shot)**
    - **Raw output:** Lange definitie: iemand die iets of iemand heel leuk of mooi vindt.
    - **Prediction:** iemand die iets of iemand heel leuk of mooi vindt.
    - **Scores:** COMET-22 = 0.9766, XCOMET = 1.0000

  - **Evaluation: Excellent.** The output is a perfect match to the reference, capturing the correct nuance. The only minor flaw is the addition of a period at the end, a slight stylistic inconsistency. Likely inherent to how the model was pre-trained.

Sample 2: ***lichtgroen*** (Triggered by high score from Aya-101)

- **Short definition:** met een lichtgroene kleur
- **Long definition:** een lichtgroene kleur hebbend

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** een lichtgroene kleur hebbend
  - **Scores:** COMET-22 = 0.9853, XCOMET = 0.9508
  - **Evaluation: Excellent.** The prediction is a perfect match to the reference.
- **Aya-101 (Few-Shot)**
  - **Prediction:** Een lichtgroene kleur hebbend.
  - **Scores:** COMET-22 = 0.9740, XCOMET = 1.0000
  - **Evaluation: Excellent.** Near-perfect match. The only deviation is the capitalization of the first letter and the addition of a period, a minor stylistic error.

Sample 3: ***zeefdruk*** (Triggered by high score from Aya-101)

- **Short definition:** afdruk door zeefdruktechniek
- **Long definition:** afdruk vervaardigd door de zeefdruktechniek

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** afdruk vervaardigd volgens de zeefdruktechniek
  - **Scores:** COMET-22 = 0.9544, XCOMET = 0.9681

  - **Evaluation: Good.** Semantically correct and stylistically appropriate. It uses "volgens" (according to) instead of "door" (by), which is a valid phrasing.
- **Aya-101 (Few-Shot)**
  - **Raw Output**: Lange definitie: afdruk vervaardigd door zeefdruktechniek
  - **Prediction:** afdruk vervaardigd door zeefdruktechniek
  - **Scores:** COMET-22 = 0.9704, XCOMET = 1.0000
  - **Evaluation: Excellent.** The prediction is a perfect lexical match to the reference, demonstrating strong adherence to the target style.

Sample 4: ***ecoroman*** (Triggered by high score from Aya-101)

- **Short definition:** roman over milieuproblematiek
- **Long definition:** roman die over milieuproblematiek of andere milieuvraagstukken gaat

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** roman waarin milieuproblematiek centraal staat
  - **Scores:** COMET-22 = 0.8813, XCOMET = 0.9449
  - **Evaluation: Good.** Semantically accurate but slightly less comprehensive than the reference. It captures the core idea but omits the explicit mention of "andere milieuvraagstukken" (other environmental issues).
- **Aya-101 (Few-Shot)**
  - **Raw Output**: Lange definitie: roman die over milieuproblematiek en andere milieuvraagstukken gaat
  - **Prediction:** roman die over milieuproblematiek en andere milieuvraagstukken gaat
  - **Scores:** COMET-22 = 0.9658, XCOMET = 1.0000

   - **Evaluation: Excellent.** A near-perfect match that correctly includes the expansion to "andere milieuvraagstukken", showing a more complete understanding of the expansion task.

Sample 5: ***muzikante*** (Triggered by high score from Aya-101)

- **Short definition:** vrouwelijke muzikant
- **Long definition:** vrouw die muziek maakt voor haar beroep of uit liefhebberij

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
   - **Prediction:** vrouw die muziek maakt, voor haar beroep of uit liefhebberij
   - **Scores:** COMET-22 = 0.9550, XCOMET = 0.9643
   - **Evaluation: Excellent.** A near-perfect match, correctly expanding the definition to include the professional or hobbyist aspect. The comma placement is slightly different but grammatically correct.
- **Aya-101 (Few-Shot)**
   - **Raw Output**: Lange definitie: vrouw die voor haar beroep of uit liefhebberij muziek maakt
   - **Prediction:** vrouw die voor haar beroep of uit liefhebberij muziek maakt
   - **Scores:** COMET-22 = 0.9652, XCOMET = 0.9983
   - **Evaluation: Excellent.** Also a near-perfect match. It slightly alters the word order ("voor haar beroep of uit liefhebberij muziek maakt") which is a valid variation.

Sample 6: ***Wageningse*** (Triggered by high score from Aya-101)

- **Short definition:** vrouw uit Wageningen
- **Long definition:** vrouw die afkomstig is uit Wageningen

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
    - **Prediction:** vrouw die afkomstig is uit Wageningen
    - **Scores:** COMET-22 = 0.9810, XCOMET = 0.9591
    - **Evaluation: Excellent.** A perfect match to the reference, correctly adding the formal term "afkomstig is uit" (originates from).
- **Aya-101 (Few-Shot)**
    - **Prediction:** vrouw die afkomstig is uit de gemeente Wageningen
    - **Scores:** COMET-22 = 0.9611, XCOMET = 1.0000
    - **Evaluation: Good. Minor Factual Hallucination.** The model adds "uit de gemeente Wageningen" (from the municipality of Wageningen). While likely true, this level of specificity is not in the reference and represents a slight, unprompted addition of information.

Sample 7: ***horeca-exploitant*** (Triggered by high score from Aya-101)

- **Short definition:** uitbater van een horecabedrijf
- **Long definition:** iemand die voor zijn beroep een horecabedrijf uitbaat

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
    - **Prediction:** iemand die voor zijn beroep een horecabedrijf uitbaat
    - **Scores:** COMET-22 = 0.9770, XCOMET = 0.9775
    - **Evaluation: Excellent.** A perfect match. It correctly identifies that the expansion requires adding the professional context "iemand die voor zijn beroep...".
- **Aya-101 (Few-Shot)**
    - **Prediction:** iemand die voor zijn beroep een horecabedrijf exploiteert
    - **Scores:** COMET-22 = 0.9592, XCOMET = 1.0000

  - **Evaluation: Good.** Semantically correct but uses the verb "exploiteert" (exploits/operates) instead of "uitbaat". While technically correct, "uitbaat" is more common and natural in this context.

Sample 8: ***beroepsvereniging*** (Triggered by high score from Aya-23)

- **Short definition:** belangenvereniging van mensen met hetzelfde beroep
- **Long definition:** vereniging van personen die hetzelfde beroep uitoefenen en gezamenlijk opkomen voor hun belangen

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** vereniging van mensen die hetzelfde beroep uitoefenen en gezamenlijk opkomen voor hun belangen
  - **Scores:** COMET-22 = 0.9662, XCOMET = 0.9910
  - **Evaluation: Excellent.** A near-perfect match, correctly expanding on the purpose of the association.
- **Aya-101 (Few-Shot)**
  - **Raw Output**: Lange definitie: vereniging van personen met hetzelfde beroep die gezamenlijk opkomen voor hun belangen
  - **Prediction:** vereniging van personen met hetzelfde beroep die gezamenlijk opkomen voor hun belangen
  - **Scores:** COMET-22 = 0.9348, XCOMET = 1.0000
  - **Evaluation: Good.** Semantically correct but slightly less precise. It uses "personen met hetzelfde beroep" instead of the more formal "personen die hetzelfde beroep uitoefenen".

Sample 9: ***groepsspreekuur*** (Triggered by high score from Aya-101)

- **Short definition:** spreekuur voor meer patiënten
- **Long definition:** spreekuur waarbij meer patiënten met een gelijksoortige, vaak chronische aandoening tegelijkertijd geholpen worden

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
    - **Prediction:** spreekuur dat wordt gehouden door een arts voor een aantal patiënten die dezelfde klacht of ziekte hebben
    - **Scores:** COMET-22 = 0.7783, XCOMET = 0.9274
    - **Evaluation: Sense Drift.** The model drifts from the core concept. It defines it as a session "gehouden door een arts" (held by a doctor) for patients with the "dezelfde klacht of ziekte" (same complaint or illness). While plausible, it misses the key reference detail of being for "gelijksoortige, vaak chronische aandoening" (similar, often chronic conditions).
- **Aya-101 (Few-Shot)**
    - **Prediction:** spreekuur waarbij meer patiënten met een gelijksoortige, vaak chronische, aandoening tegelijkertijd geholpen worden
    - **Scores:** COMET-22 = 0.9590, XCOMET = 0.9981
    - **Evaluation: Excellent.** A near-perfect match, correctly capturing all the key semantic components from the reference, including the important detail about chronic conditions.

Sample 10: *lifestylecoach* (Triggered by high score from Aya-101)

- **Short definition:** coach voor lifestyle
- **Long definition:** iemand die mensen advies geeft op het gebied van voeding en beweging

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
    - **Prediction:** coach die mensen adviseert en begeleidt op het gebied van hun leefstijl
    - **Scores:** COMET-22 = 0.8382, XCOMET = 0.8794
    - **Evaluation: Sense Drift.** The definition "coach die mensen adviseert en begeleidt op het gebied van hun leefstijl" (coach who advises and guides people in the area of their lifestyle) is circular and doesn't

provide the specific details (nutrition and exercise) that are in the reference.

- **Aya-101 (Few-Shot)**
  - **Raw Output**: Lange definitie: iemand die mensen advies geeft op het gebied van voeding en beweging.
  - **Prediction:** iemand die mensen advies geeft op het gebied van voeding en beweging.
  - **Scores:** COMET-22 = 0.9608, XCOMET = 0.9951
  - **Evaluation: Excellent.** A perfect match to the reference, correctly identifying that "lifestyle" in this context expands to "voeding en beweging" (nutrition and exercise).

Sample 11: *meisjesmeisje* (Triggered by high score from Aya-101)

- **Short definition:** zeer meisjesachtig meisje
- **Long definition:** meisje dat zich zeer meisjesachtig gedraagt en kleedt

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** meisje dat zeer meisjesachtig is
  - **Scores:** COMET-22 = 0.6646, XCOMET = 0.9105
  - **Evaluation: Semantic Omission.** The definition "meisje dat zeer meisjesachtig is" is a tautology and fails to expand on the key behaviors of "gedraagt en kleedt" (behaves and dresses) from the reference.
- **Aya-101 (Few-Shot)**
  - **Prediction:** Een meisje dat zich zeer meisjesachtig gedraagt en kleedt.
  - **Scores:** COMET-22 = 0.9583, XCOMET = 0.9973
  - **Evaluation: Excellent.** A near-perfect match that correctly captures the behavioral and stylistic aspects of the definition.

Sample 12: ***eenoudergezin*** (Triggered by high score from Aya-23)

- **Short definition:** gezin met één ouder
- **Long definition:** gezin dat bestaat uit één ouder met een of meer kinderen

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** gezin dat bestaat uit één ouder en een of meer kinderen
  - **Scores:** COMET-22 = 0.9694, XCOMET = 0.9847
  - **Evaluation: Excellent.** A near-perfect match, correctly adding the essential component "en een of meer kinderen".
- **Aya-101 (Few-Shot)**
  - **Prediction:** gezin dat bestaat uit één ouder en een of meer kinderen
  - **Scores:** COMET-22 = 0.9727, XCOMET = 1.0000
  - **Evaluation: Excellent.** Also a near-perfect match to the reference.

Sample 13: ***roetschade*** (Triggered by high score from Aya-23)

- **Short definition:** schade door roet
- **Long definition:** schade die veroorzaakt wordt door roet

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** schade die wordt veroorzaakt door roet
  - **Scores:** COMET-22 = 0.9762, XCOMET = 0.9778
  - **Evaluation: Excellent.** A near-perfect match.
- **Aya-101 (Few-Shot)**
  - **Raw Output**: Lange definitie: schade veroorzaakt door roet.
  - **Prediction:** schade veroorzaakt door roet.
  - **Scores:** COMET-22 = 0.9533, XCOMET = 1.0000

  - **Evaluation: Good.** Semantically correct, but "schade veroorzaakt door roet" is slightly less fluent and lexicographically formal than the reference.

Sample 14: ***poëziewedstrijd*** (Triggered by high score from Aya-23)

- **Short definition:** wedstrijd in het schrijven van gedichten
- **Long definition:** wedstrijd waarbij de deelnemers een gedicht schrijven en een jury bepaalt wie daarin het beste is geslaagd

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** wedstrijd waarbij de deelnemers een gedicht schrijven en een jury bepaalt wie daarin het beste geslaagd is
  - **Scores:** COMET-22 = 0.9695, XCOMET = 0.9736
  - **Evaluation: Excellent.** A perfect match, correctly expanding the definition to include the concept of a jury.
- **Aya-101 (Few-Shot)**
  - **Raw Output**: Lange definitie: poëziewedstrijd is een wedstrijd in het schrijven van gedichten.
  - **Prediction:** poëziewedstrijd is een wedstrijd in het schrijven van gedichten.
  - **Scores:** COMET-22 = 0.7641, XCOMET = 0.9431
  - **Evaluation: Semantic Omission / Stylistic Error.** The model simply repeats the input in a full sentence ("poëziewedstrijd is een wedstrijd..."), failing to add the crucial information about the jury. This is not a valid expansion.

Sample 15: ***trouwplanner*** (Triggered by high score from Aya-23)

- **Short definition:** iemand die bruiloften organiseert
- **Long definition:** iemand die zich voor zijn beroep bezighoudt met de organisatie van bruiloften

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** iemand die zich voor zijn beroep bezighoudt met het organiseren van bruiloften
  - **Scores:** COMET-22 = 0.9607, XCOMET = 0.9799
  - **Evaluation: Excellent.** A near-perfect match, correctly adding the professional context.
- **Aya-101 (Few-Shot)**
  - **Prediction:** Een trouwplanner is iemand die bruiloften organiseert.
  - **Scores:** COMET-22 = 0.7330, XCOMET = 1.0000
  - **Evaluation: Semantic Omission / Stylistic Error.** This is a non-expansion that just rephrases the input into a sentence. It completely misses the "for his profession" aspect.

Sample 16: ***staalproduct*** (Triggered by high score from Aya-23)

- **Short definition:** product van staal
- **Long definition:** product waarin staal de belangrijkste grondstof is

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** product waarvan staal de belangrijkste grondstof is
  - **Scores:** COMET-22 = 0.9716, XCOMET = 0.9685
  - **Evaluation: Excellent.** A near-perfect match, correctly identifying the key expansion is to specify that steel is the "belangrijkste grondstof" (most important raw material).
- **Aya-101 (Few-Shot)**
  - **Prediction:** Product waarin staal de belangrijkste grondstof is
  - **Scores:** COMET-22 = 0.9818, XCOMET = 1.0000
  - **Evaluation: Excellent.** A perfect match.

Sample 17: ***benoemingstermijn*** (Triggered by high score from Aya-23)

- **Short definition:** termijn waarvoor iemand benoemd wordt
- **Long definition:** termijn waarvoor iemand in een bepaalde functie benoemd wordt

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** termijn waarvoor iemand voor een functie of ambt wordt benoemd
  - **Scores:** COMET-22 = 0.9562, XCOMET = 0.9804
  - **Evaluation: Excellent.** The expansion to "voor een functie of ambt" (for a function or office) is a very lexicographically appropriate addition.
- **Aya-101 (Few-Shot)**
  - **Raw Output**: Lange definitie: termijn waarvoor iemand benoemd wordt
  - **Prediction:** termijn waarvoor iemand benoemd wordt
  - **Scores:** COMET-22 = 0.9632, XCOMET = 0.9854
  - **Evaluation: Semantic Omission.** A complete failure of the expansion task; the output is identical to the input.

Sample 18: ***dorpswinkelier*** (Triggered by high score from Aya-23)

- **Short definition:** uitbater van een dorpswinkel
- **Long definition:** iemand die voor zijn beroep de dorpswinkel uitbaat

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** iemand die voor zijn beroep een dorpswinkel uitbaat
  - **Scores:** COMET-22 = 0.9581, XCOMET = 0.9766
  - **Evaluation: Excellent.** A perfect match, correctly adding the professional context.

- **Aya-101 (Few-Shot)**
  - **Prediction:** iemand die voor zijn beroep of als vrijwilliger een dorpswinkel uitbaat
  - **Scores:** COMET-22 = 0.9227, XCOMET = 0.9997
  - **Evaluation: Minor Factual Hallucination.** The addition of "of als vrijwilliger" (or as a volunteer) is not present in the reference. While a village shop could be run by volunteers, this is an unprompted addition of information.

Sample 19: ***aspergerobot*** (Triggered by high score from Aya-23)

- **Short definition:** machine om asperges te oogsten
- **Long definition:** machine die gebruikt wordt om asperges te oogsten

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** machine die wordt gebruikt om asperges te oogsten
  - **Scores:** COMET-22 = 0.9661, XCOMET = 0.9683
  - **Evaluation: Excellent.** A perfect match.
- **Aya-101 (Few-Shot)**
  - **Prediction:** Aspergerobot: machine om asperges te oogsten
  - **Scores:** COMET-22 = 0.8457, XCOMET = 0.9385
  - **Evaluation: Stylistic error and semantic omission.** The model prepends the lemma ("Aspergerobot:") to the definition, which is a significant formatting error. The definition itself is a non-expansion.

Sample 20: ***kleutertuin*** (Triggered by high score from Aya-23)

- **Short definition:** gebouw van een kleutertuin
- **Long definition:** gebouw waarin een kleutertuin is gehuisvest

Model outputs and evaluation

- **Aya-23 (Fine-Tuned)**
  - **Prediction:** gebouw waarin een kleutertuin gehuisvest is
  - **Scores:** COMET-22 = 0.9653, XCOMET = 0.9690
  - **Evaluation: Excellent.** A near-perfect match, correctly expanding to specify the building "is gehuisvest" (is housed).
- **Aya-101 (Few-Shot)**
  - **Prediction:** Een kleutertuin is een kinderdagverblijf of peuterspeelzaal voor kinderen van 0 tot 5 jaar.
  - **Scores:** COMET-22 = 0.7124, XCOMET = 0.9687
  - **Evaluation: Major sense drift and factual hallucination.** The model provides a completely incorrect definition, confusing "kleutertuin" (kindergarten/preschool, an older term) with "kinderdagverblijf of peuterspeelzaal" (daycare or playgroup). This is a critical semantic failure.

A.2 Genus proxima groups

This appendix presents the detailed qualitative analysis of the selected genus proxima groups. The analysis assessed the models' ability to maintain structural and stylistic consistency across related lexical items. Table 1: The evaluation grid for Dutch definition expansions. was used to accomplish this.

Genus proxima: ***liefhebber*** (enthusiast/lover of)

Lemma: ***operaliefhebber***

- **Short definition**: liefhebber van operamuziek
- **Long definition**: iemand die veel van het spelen van en/of het luisteren naar operamuziek houdt

Output from Aya-23 (Fine-Tuned):

- **Prediction**: iemand die veel van het zingen van opera en het bijbehorende muziekgenre houdt

- **Scores:** COMET-22 = 0.8390, XCOMET = 0.7690
- **Evaluation: Minor sense drift**. The model correctly identifies the structure but specifies "zingen van opera" (singing of opera), which is narrower than the reference's "spelen van en/of luisteren naar" (playing and/or listening to).

Output from Aya-101 (Few-Shot):

- **Prediction**: iemand die van operamuziek houdt
- **Scores**: COMET-22 = 0.8667, XCOMET = 0.9942
- **Evaluation: Semantic omission.** A complete failure of the expansion task; the output barely differentiates from the input.
- **Few-Shot Example Analysis**: The examples provided were highly relevant. klassiekemuziekliefhebber is a perfect parallel example, sharing the exact structure of the reference. The model failed to fully replicate this structure, suggesting a limitation in applying the provided pattern.

Lemma: *sportliefhebber*

- **Short definition**: liefhebber van sport
- **Long definition**: iemand die graag sport en/of veel van het kijken naar sport houdt

Output from Aya-23 (Fine-Tuned):

- **Prediction**: iemand die veel van het beoefenen of volgen van sport houdt
- **Scores**: COMET-22 = 0.8610, XCOMET = 0.9684
- **Evaluation: Good**. A high-quality paraphrase of the reference. "Beoefenen of volgen" (practicing or following) is a very natural and accurate way to express the concept.

Output from Aya-101 (Few-Shot):

- **Prediction**: Een sportliefhebber is iemand die van sport houdt.
- **Scores**: COMET-22 = 0.8543, XCOMET = 1.0000
- **Evaluation: Stylistic error and omission**. This is a non-expansion that simply rephrases the input into a sentence. It fails to add the crucial "practicing and/or watching" distinction.

- **Few-Shot Example Analysis**: The examples (sportlustige, sportgek) were thematically related but did not provide the correct structural pattern for expansion. This likely contributed to the model's failure to produce a detailed definition.

Lemma: *taalliefhebber*

- **Short definition**: liefhebber van taal
- **Long definition**: iemand die veel van taal houdt en daar graag over spreekt en leest

Output from Aya-23 (Fine-Tuned):

- **Prediction**: iemand die veel van taal en taalgebruik houdt
- **Scores**: COMET-22 = 0.8093, XCOMET= 0.9128
- **Evaluation: Good**. The expansion to include "taalgebruik" (language use) is valid and good, but it omits the reference's detail about "spreekt en leest" (speaks and reads).

Output from Aya-101 (Few-Shot):

- **Prediction**: iemand die van taal houdt.
- **Scores**: COMET-22 = 0.7481, XCOMET = 0.9962
- **Evaluation: Omission**. Another minimal expansion that fails to capture any of the specific details from the reference definition.
- **Few-Shot Example Analysis**: The examples (taalkundige, afasie) were topically relevant to language but structurally useless for defining a type of person ("liefhebber"). This lack of a structural analogue likely led to the poor output.

Lemma: *theaterliefhebber*

- **Short definition**: liefhebber van theater
- **Long definition**: iemand die veel van het spelen en/of bijwonen van theatervoorstellingen houdt

Output from Aya-23 (Fine-Tuned):

- **Prediction**: iemand die veel van het bezoeken van het theater houdt
- **Scores**: COMET-22 = 0.8519, XCOMET = 0.9544
- **Evaluation: Good**. The definition is good but less complete than the reference. It captures "bijwonen" (attending) but omits the "spelen" (acting/playing) aspect.

Output from Aya-101 (Few-Shot):

- **Prediction**: iemand die veel van het kijken naar en/of spelen in toneelstukken houdt
- **Scores**: COMET-22 = 0.9154, XCOMET = 0.8755
- **Evaluation: Excellent**. A very strong output. It correctly identifies the "spelen" and "kijken naar" (watching) components. "Toneelstukken" (plays) is a reasonable synonym for "theatervoorstellingen".
- **Few-Shot Example Analysis**: The model received a perfect structural analogue with the example toneelliefhebber. It successfully identified and replicated this pattern, demonstrating a strong ability to generalize when given a good example.

Genus proxima: *ologie* (the study of)

Lemma: *hematologie*

- **Short definition**: medisch specialisme voor bloedziekten
- **Long definition**: medisch specialisme dat zich bezighoudt met het onderzoek, de diagnose en behandeling van bloedziekten, en aandoeningen aan de bloedvormende organen en de lymfeklieren

Output from Aya-23 (Fine-Tuned):

- **Prediction**: medisch specialisme dat zich bezighoudt met de diagnose, behandeling en het onderzoek van bloedziekten en bloedstoornissen
- **Scores**: COMET-22 = 0.8857, XCOMET = 0.8990
- **Evaluation: Good**. The model correctly expands to include "diagnose, behandeling en het onderzoek" but omits the second part of the reference concerning "bloedvormende organen en de lymfeklieren".

Output from Aya-101 (Few-Shot):

- **Prediction**: Hematologie is een medisch specialisme dat zich bezighoudt met de diagnose, behandeling en onderzoek van bloedziekten en aandoeningen aan de bloedvormende organen en de lymfeklieren.
- **Scores**: COMET-22 = 0.9293, XCOMET = 0.9946
- **Evaluation: Excellent**. Semantically, this is a near-perfect match to the reference. The only issue is the stylistic choice to formulate it as a full sentence ("Hematologie is...").
- **Few-Shot Example Analysis**: The model was provided with hematoloog and reumatologie, both of which are excellent structural examples. It successfully leveraged these to produce a comprehensive and accurate definition.

Lemma: *nefrologie*

- **Short definition**: medisch specialisme voor nierziekten
- **Long definition**: medisch specialisme dat zich bezighoudt met het onderzoek, de diagnose en behandeling van nierziekten

Output from Aya-23 (Fine-Tuned):

- **Prediction**: medisch specialisme dat zich bezighoudt met de diagnose, behandeling en het onderzoek van nierziekten
- **Scores**: COMET-22 = 0.9639, XCOMET = 0.9703
- **Evaluation: Excellent**. A near-perfect match, with only a minor reordering of "onderzoek, diagnose en behandeling".

Output from Aya-101 (Few-Shot):

- **Prediction**: medisch specialisme dat zich bezighoudt met het onderzoek, de diagnose en behandeling van nierziekten
- **Scores**: COMET-22 = 0.9766, XCOMET = 1.0000
- **Evaluation: Excellent**. A perfect match to the reference.
- Few-Shot Example Analysis: The examples nefroloog and reumatologie provided perfect structural patterns. The model correctly identified and applied this pattern, resulting in a flawless output.

Lemma: ***pneumologie***

- **Short definition**: medisch specialisme voor longziekten
- **Long definition**: medisch specialisme dat zich bezighoudt met het onderzoek, de diagnose en behandeling van longziekten

Output from Aya-23 (Fine-Tuned):

- **Prediction**: medisch specialisme dat zich bezighoudt met de diagnose, behandeling en het onderzoek van longziekten en ademhalingsproblemen
- **Scores**: COMET-22 = 0.9238, XCOMET = 0.9297
- **Evaluation: Good**. The model correctly identifies the core structure but adds "en ademhalingsproblemen" (and respiratory problems). While related, this is not in the reference and constitutes a minor, unprompted addition of information.

Output from Aya-101 (Few-Shot):

- **Prediction**: Pneumologie is het medisch specialisme voor longziekten.
- **Scores**: COMET-22 = 0.7562, XCOMET = 1.0000
- **Evaluation: Omission and stylistic Error**. This is a non-expansion that simply reformulates the input as a sentence. It fails to add the key details about "onderzoek, de diagnose en behandeling".
- **Few-Shot Example Analysis**: The examples provided (longstayer, pneumonie) were thematically relevant but structurally poor. Neither provided the "-logie" definition pattern seen in the other successful examples. This lack of a good structural prompt likely caused the model to fall back on a simple, unhelpful reformulation.

B Generative AI use disclaimer

GenAI code of conduct for students (2024-2025)

*Generative AI (GenAI) assistance tools can be used to generate text, image, code, video, music or combinations of these***.** *It includes typical tools like (but this list is not limited to): ChatGPT, Google Gemini, MS Copilot, Midjourney, Claude.ai, Perplexity.ai, Dall-E, …*

Student name: Robbe Meersman

Student number: r0718552

Please indicate with "X" whether it relates to a course assignment or to the master thesis:

O This form is related to a **course assignment**.

Course name: ....

Course number: ....

X This form is related to **my Master thesis**.

Title Master thesis: Transformer-based Expansion of Dutch Dictionary Definitions

Promoter: Prof. Dr. Vincent Vandeghinste

Daily supervisor: Prof. Dr. Kris Heylen

Please indicate with "X":

O **I did not use** any GenAI assistance tool.

X I did use GenAI Assistance. In this case **specify which ones** (e.g. ChatGPT, ...):

	Name of the GenAI tool used *Please indicate with "X" (possibly multiple times) in which way you were using GenAI:*		
GenAI assistance used as/for:	Gemini 2.5 pro	Chat GPT o3	Chat GPT 4.5
Language assistance	X		X
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Short-form input assistance			
Generating programming code	X	X	
Generating new research ideas			
Generating blocks of text			
Other (specify):			

X As a language assistant for reviewing or improving texts I wrote myself

- ➢ *Code of conduct*: This use is similar to using spelling and grammar check tools, you do not have to refer to use of GenAI in the text.
  Be careful:
  - o Using GenAI tools on texts you did not write yourself to cover up plagiarism (by paraphrasing original texts) is not allowed.

X As a search engine to get information on a topic or to search for existing research on the topic

- ➢ *Code of conduct*: This use is similar to e.g. a Google search or checking Wikipedia. If you then write your own text based on this information, you do not have to refer to use of GenAI in the text.
  Be careful:
  - o Be aware that the output of the GenAI tool cannot be guaranteed as a 100% reliable source of information. The output may not be entirely correct and be limited due to the databases it uses. Knowledge evolves and may change over time, it may be that the database of the GenAI tool is not up to date.

X For literature search

- ➢ *Code of conduct*: This use is comparable to e.g. a Google Scholar search.
  Be careful:
  - o Be aware that the output is restricted to the database it is built on. After this initial search, look for scientific sources and conduct your own analysis of the source documents. Interpret, analyse and process the information you obtained; verify it and don't just copy-paste it.
  - o Be aware that some GenAI tools (like ChatGPT) may output no or wrong references. As a student you are responsible for further checking and verifying the absence or correctness of references; don't just copy-paste it.

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O **For short-form input assistance**

- ➢ *Code of conduct*: This use is similar to e.g. Google docs powered by generative language models

X To generate programming code

- ➢ *Code of conduct*: Use of GenAI for coding should be explicitly allowed by the teacher. If used for coding, correctly mention the use of GenAI assistance and cite it.

O **To generate new research ideas**

- ➢ *Code of conduct*: Further verify in this case whether the idea is novel or not. It is likely that it is related to existing work, which should be referenced then.

O **To generate blocks of text**

- ➢ *Code of conduct*: Inserting blocks of text without quotes and a reference to GenAI assistance in your report or thesis is not allowed. According to Article 84 of the exam regulations in evaluating your work one should be able to correctly judge on your own knowledge, understanding and skills. In case it is really needed to insert a block of text from a GenAI tool, mention it as a citation by using quotes. But this should be kept to an absolute minimum. When you literally copy elements from a conversation with a GenAI tool: Quote between quotation marks and refer according to the specified reference style or as a personal communication within the text itself. Describe the use of the GenAI-tool (tool name, version, date, ...) in the method section (if there is one) and optionally add the (link to the) full conversation as an attachment.

O **Other**

- ➢ *Code of conduct*: Contact the professor of the course or the promoter of the thesis. Inform also the program director. Motivate how you comply with Article 84 of the exam regulations. Explain the use and the added value of ChatGPT or another AI tool: ….

Further important guidelines and remarks

- GenAI assistance cannot be used related **to data or subjects under Non-Disclosure Agreement.**
- GenAI assistance cannot be used related **to sensitive or personal data due to privacy issues**.
- **Take a scientific and critical attitude** when interacting with GenAI assistance and interpreting its output. Don't become emotionally connected to AI tools.
- As a student you are responsible for complying with Article 84 of the exam regulations: your report or thesis should reflect your own knowledge, understanding and skills. Be aware that plagiarism rules also apply to (work that is the result of) the use of GenAI assistance tools.
- **Exam regulations Article 84**: "Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge, understanding and/or skills or those of other students, is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e. copying the work (ideas, texts, structures, designs, images, plans, codes , ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources. Every possession of prohibited resources during an examination (see article 65) is considered an irregularity."
- In order to maintain academic integrity and avoid plagiarism **more information about being transparent on the use of GenAI assistance and about citing and referencing GenAI** can be found on this website for students (Dutch/English).
- **Additional reading : KU Leuven guidelines on responsible use of Generative AI tools, and other information** (Dutch/English)

A few final words

If you are uncertain whether or not you should declare your use of AI tools, we suggest that you discuss the matter with your teacher or promoter. It is safer to declare AI use when it is not needed than to withhold that declaration when it is required.

Finally, remember that advanced AI tools are new and that they can do things they could not do, up until recently – so we do not have all the answers about how to use them responsibly yet. It is important to follow-up on most recent evolutions in AI technologies, to be a bit cautious, to communicate with teachers, teachings assistants, supervisors, promoters and peers with open minds, to be as transparent as we can, and to learn together as we move along.

This code of conduct can be used as a framework for academic year 2024-2025, changes will be made based on new evolutions.

Bibliography

[1] Thanapon Noraset, Chen Liang, Larry Birnbaum and Doug Downey, "Definition modeling: Learning to define word embeddings in natural language," in *AAAI Conference on Artificial Intelligence (AAAI-17)*, San Francisco, CA, USA, 2017.

[2] Artyom Gadetsky, Ilya Yakubovskiy and Dmitry Vetrov, "Conditional generators of word definitions," in *ACL 2018 (Short Papers), 56th Annual Meeting of the Association for Computational Linguistics*, Melbourne, Australia, 2018.

[3] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser and Illia Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems 30*, Long Beach, CA, USA, 2017.

[4] R. Lew, "ChatGPT as a COBUILD lexicographer," *Humanities and Social Sciences Communications,* vol. 10, 2023.

[5] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li and Peter J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," *Journal of Machine Learning Research,* vol. 21, no. 140, pp. 1-67, 2020.

[6] Christoph Leiter, Piyawat Lertvittayakumjorn, Marina Fomicheva, Wei Zhao, Yang Gao and Steffen Eger, "Towards explainable evaluation metrics for machine translation," *Journal of Machine Learning Research,* vol. 25, pp. 1-49, 2024.

[7] G. A. Miller, R. Beckwith, C. Fellbaum, D. Gross and K. Miller, "Introduction to WordNet: An On-line Lexical Database," 1993.

[8] "Wordnet Princeton," [Online]. Available: https://wordnet.princeton.edu/.

[9] G. A. Miller, "WordNet: a lexical database for English," *Communications of the ACM,* vol. 38, no. 11, pp. 39-41, 1995.

[10] H. G. Oliveira and P. Gomes, "Towards the Automatic Creation of a Wordnet from a Term-based Lexical Network," in *Proceedings of the 2010 Workshop on Graph-based Methods for Natural Language Processing, ACL*, Uppsala, Sweden, 2010.

[11] M. Sahlgren, "The distributional hypothesis*," Kista, Sweden, 2008.

[12] Tomas Mikolov, Kai Chen, Gregory S. Corrado and Jeffrey Dean, "Efficient Estimation of Word Representations in Vector Space," in *1st International Conference on Learning Representations (ICLR 2013), Worksh op Track Proceedings*, Scottsdale, AZ, 2013.

[13] Tomas Mikolov, Ilya Sutskever, Kai Chen, Gregory S. Corrado and Jeffrey Dean, "Distributed Representations of Words and Phrases and their Compositionality," in *Advances in Neural Information Processing Systems 26 (NIPS 2013)*, Lake Tahoe, NV, 2013.

[14] Jeffrey Pennington, Richard Socher and Christopher D. Manning, "GloVe: Global Vectors for Word Representation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP 2014)*, Doha, Qatar, 2014.

[15] F. Hill, K. Cho, A. Korhonen and Y. Bengio, "Learning to Understand Phrases by Embedding the Dictionary," *Transactions of the Association for Computational Linguistics,* vol. 4, pp. 17-30, 2016.

[16] J. Fodor, S. D. Deyne and S. Suzuki, "The Importance of Context in the Evaluation of Word Embeddings: The Effects of Antonymy and Polysemy," in *Proceedings of the 15th International Conference on Computational Semantics*, Nancy, France, 2023.

[17] Matthew E. Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee and Luke Zettlemoyer, "Deep contextualized word representations," in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, New Orleans, LA, 2018.

[18] Jacob Devlin, Ming-Wei Chang, Kenton Lee and Kristina Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding,"

in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Minneapolis, MN, 2019.

[19] Dzmitry Bahdanau, Kyunghyun Cho and Yoshua Bengio, "Neural Machine Translation by Jointly Learning to Align and Translate," in *3rd International Conference on Learning Representations (ICLR 2015), Conference Track Proceedings*, San Diego, CA, 2015.

[20] A. Rogers, O. Kovaleva and A. Rumshisky, "A Primer in BERTology: What We Know About How BERT Works," *Transactions of the Association for Computational Linguistics,* vol. 8, pp. 842-866, 2020.

[21] Rajesh Gupta, "Bidirectional encoders to state-of-the-art: a review of BERT and its transformative impact on natural language processing," *Informatics. Economics. Management,* vol. 4, no. 1, p. 311 – 320, 2024.

[22] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan and R. Child, "Language Models are Few-Shot Learners," in *Advances in Neural Information Processing Systems 33 (NeurIPS)*, Online, 2020.

[23] L. Xue, N. Constant, A. Roberts, M. Kale, R. Al-Rfou, A. Siddhant, A. Barua and C. Raffel, "mT5: A Massively Multilingual Pre-trained Text-to-Text Transformer," in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Online, 2021.

[24] B. Vanroy, "GEITje 7B Ultra: A Conversational Model for Dutch," arXiv:2412.04092 [cs.CL], Leuven, Belgium, 2024.

[25] Rijgersberg, "GEITje README," Github, January 2025. [Online]. Available: https://github.com/Rijgersberg/GEITje/blob/main/README-en.md. [Accessed July 2025].

[26] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning and C. Finn, "Direct Preference Optimization: Your Language Model is Secretly a Reward Model," arXiv:2305.18290v3 [cs.LG], 2024.

[27] A. Üstün, V. Aryabumi, Z. Yong, W.-Y. Ko, D. D'souza, G. Onilude, N. Bhandari, S. Singh, H.-L. Ooi, A. Kayid, F. Vargus, P. Blunsom, S. Longpre, N. Muennighoff,

M. Fadaee, J. Kreutzer and S. Hooker, “Aya Model: An Instruction Finetuned Open-Access Multilingual Language Model,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, Bangkok, Thailand, 2024.

[28] alexrs, “Huggingface,” [Online]. Available: https://huggingface.co/CohereLabs/aya-101/blob/main/README.md. [Accessed July 2025].

[29] V. Aryabumi, J. Dang, D. Talupuru, S. Dash, D. Cairuz, H. Lin, B. Venkitesh, M. Smith, J. A. Campos, Y. C. Tan, K. Marchisio, M. Bartolo, S. Ruder, A. Locatelli, J. Kreutzer, N. Frosst and A. Gomez, “Aya 23: Open Weight Releases to Further Multilingual Progress,” arXiv:2405.15032v2 [cs.CL], 2024.

[30] P. Delobelle and F. Remy, “RobBERT-2023: Keeping Dutch Language Models Up-To-Date at a Lower Cost Thanks to Model Conversion,” *Computational Linguistics in the Netherlands Journal,* vol. 13, pp. 193-203, 2024.

[31] “GPT-NL,” [Online]. Available: https://gpt-nl.nl/. [Accessed July 2025].

[32] “A language model built for the public good,” ETH Zürich, 9 July 2025. [Online]. Available: https://ethz.ch/en/news-and-events/eth-news/news/2025/07/a-language-model-built-for-the-public-good.html. [Accessed 24 July 2025].

[33] L. Wang, S. Chen, L. Jiang, S. Pan, R. Cai, S. Yang and F. Yang, “Parameter-Efficient Fine-Tuning in Large Models: A Survey of Methodologies,” arXiv:2410.19878v3 [cs.CL], 2025.

[34] Edward J. Hu, Yelong Shen, . Wallis, . Allen-Zhu, . Li, . Wang, . Wang and . Chen, “LoRA: Low-rank adaptation of large language models,” 2021.

[35] B. Jacob, S. Kligys, B. Chen, M. Zhu, M. Tang, A. Howard, H. Adam and D. Kalenichenko, “Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR),* pp. 2704-2713, 2018.

[36] T. Dettmers, A. Pagnoni, A. Holtzman and L. Zettlemoyer, “QLORA: Efficient Finetuning of Quantized LLMs,” arXiv:2305.14314 [cs.LG], 2023.

[37] B. Lester, R. Al-Rfou and N. Constant, “The Power of Scale for Parameter-Efficient Prompt Tuning,” in *Proceedings of the 2021 Conference on Empirical*

Methods in Natural Language Processing, Online and Punta Cana, Dominican Republic, 2021.

[38] X. L. Li and P. Liang, “Prefix-Tuning: Optimizing Continuous Prompts for Generation,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, Online, 2021.

[39] E. B. Zaken, Y. Goldberg and S. Ravfogel, “BitFit: Simple Parameter-efficient Fine-tuning for Transformer-based Masked Language-models,” *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics,* vol. 2, pp. 1-9, 2022.

[40] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. d. Laroussilhe, A. Gesmundo, M. Attariyan and S. Gelly, “Parameter-Efficient Transfer Learning for NLP,” arXiv:1902.00751v2 [cs.LG] , 2019.

[41] J. Pfeiffer, I. Vulić, I. Gurevych and S. Ruder, “MAD-X: An Adapter-Based Framework for Multi-Task Cross-Lingual Transfer,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Online, 2020.

[42] M. Freitag, N. Mathur, D. Deutsch, C.-K. Lo, E. Avramidis, R. Rei, B. Thompson, F. Blain, J. W. Tom Kocmi, D. I. Adelani, M. Buchicchio, C. Zerva and A. Lavie, “Are LLMs Breaking MT Metrics? Results of the WMT24 Metrics Shared Task,” in *Proceedings of the Ninth Conference on Machine Translation*, Miami, Florida, USA, 2024.

[43] K. Papineni, S. Roukos, T. Ward and W.-J. Zhu, “BLEU: a Method for Automatic Evaluation of Machine Translation,” in *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, Philadelphia, Pennsylvania, USA, 2002.

[44] “Metric: bleu,” Huggingface, [Online]. Available: https://huggingface.co/spaces/evaluate-metric/bleu. [Accessed July 2025].

[45] C.-Y. Lin, “ROUGE: A Package for Automatic Evaluation of Summaries,” *Association for Computational Linguistics,* vol. Text Summarization Branches Out, pp. 74-81, 2004.

[46] R. Rei, J. G. C. d. Souza, D. Alves, C. Zerva, A. C. Farinha, T. Glushkova, A. Lavie, L. Coheur and A. F. T. Martins, “COMET-22: Unbabel-IST 2022 Submission for

the Metrics Shared Task," in *Proceedings of the Seventh Conference on Machine Translation (WMT)*, Abu Dhabi, United Arab Emirates, 2022.

[47] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger and Y. Artzi, "BERTScore: Evaluating Text Generation with BERT," in *The Eighth International Conference on Learning Representations (ICLR)*, Online, 2020.

[48] N. M. Guerreiro, R. Rei, D. v. Stigt, L. Coheur, P. Colombo and A. F. T. Martins, "xCOMET: Transparent Machine Translation Evaluation through," *Transactions of the Association for Computational Linguistics,* vol. 12, pp. 979-995, 2024.

[49] S. Evert, C. Ganslmayer and C. Rink, "Multi-level analysis as a systematic Approach to evaluating the quality of AI-generated dictionary entries," in *Proceedings of the XXI EURALEX International Congress*, Cavtat, Croatia, 2024.

[50] "Over het ANW," Instituut voor Nederlandse taal, [Online]. Available: https://anw.ivdnt.org/about. [Accessed July 2025].

[51] X. Li, K. Lv, H. Yan, T. Lin, W. Zhu, Y. Ni, G. Xie, X. Wang and X. Qiu, "Unified Demonstration Retriever for In-Context Learning," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Toronto, Canada, 2023.

[52] L. Wang, N. Yang and F. Wei, "Learning to Retrieve In-Context Examples for Large Language Models," in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, St. Julian's, Malta, 202.

[53] Y. Mu, A. Reheman, Z. Cao, Y. Fan, B. Li, Y. Li, T. Xiao, C. Zhang and J. Zhu, "Augmenting Large Language Model Translators via Translation Memories," in *Findings of the Association for Computational Linguistics*, Toronto, Canada, 2023.

[54] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni and P. Liang, "Lost in the Middle: How Language Models Use Long Contexts," *Transactions of the Association for Computational Linguistics,* vol. 12, pp. 153-173, 2024.

[55] "Experiment tracking," CoreWeave, [Online]. Available: https://wandb.ai/site/experiment-tracking/. [Accessed July 2025].

[56] B. Vanroy, “GEITje 7B Ultra,” Huggingface, [Online]. Available: https://huggingface.co/BramVanroy/GEITje-7B-ultra. [Accessed July 2025].

[57] A. Wang, A. Singh, J. Michael, F. Hill, O. Levy and S. Bowman, “GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding,” in *Proceedings of the 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, Brussels, Belgium, 2018.

[58] P. Rajpurkar, J. Zhang, K. Lopyrev and P. Liang, “SQuAD: 100,000+ Questions for Machine Comprehension of Text,” in *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, Austin, Texas, 2016.

[59] R. Rei, M. Treviso, N. M. Guerreiro, C. Zerva, A. C. Farinha, C. Maroti, J. G. C. d. Souza, T. Glushkova, D. Alves, L. Coheur, A. Lavie and A. F. T. Martins, “CometKiwi: IST-Unbabel 2022 Submission for the Quality Estimation Shared Task,” in *Proceedings of the Seventh Conference on Machine Translation (WMT)*, Abu Dhabi, United Arab Emirates, 2022.

[60] B. Vanroy, “GEITje 7B Ultra Sft,” Dataloop, Januari 2025. [Online]. Available: https://dataloop.ai/library/model/bramvanroy_geitje-7b-ultra-sft/. [Accessed July 2025].